

Deceptive Agents in Extended Reality: A Speculative User Study

PETER TASKER, University of Strathclyde, Scotland

AMEL BOURDOUCEN, F-Secure, Finland

PEJMAN SAEAGHE, University of Strathclyde, Scotland

Virtual Embodied Agents (VEAs)—autonomous, goal-directed AI embodied within Extended Reality (XR)—are reshaping Deceptive Design (DD): rather than a static deceptive interface, a VEA can act on a user’s behalf, be experienced as a social actor, and operate within the user’s perceptual and spatial environment simultaneously. This introduces a new interdisciplinary dimension to DD research. We asked users to speculate on deceptive interactions with VEAs across six settings—commerce, museum tour-guidance, therapy, training, conversation, and information retrieval—and on mitigation strategies. From their responses, we develop a taxonomy of anticipated deceptive patterns organised around agentic control, social embodiment, and spatial control, alongside enabling AI/XR properties and preferred mitigations. We compare this against existing DD, XR-manipulation, and AI dark-pattern taxonomies, questioning whether current terminology captures this convergence. We show how the agent itself can be a deceptive vector, how XR amplifies deceptive effects, and discuss anthropomorphism, societal risk, and directions for mitigation research.

CCS Concepts: • **Computing methodologies** → **Artificial intelligence**; • **Security and privacy** → *Human and societal aspects of security and privacy*; • **Human-centered computing** → **Human computer interaction (HCI)**; **Mixed / augmented reality**; **Virtual reality**.

Additional Key Words and Phrases: dark patterns, deceptive design, embodied AI, virtual reality, extended reality, human-computer interaction, artificial intelligence

ACM Reference Format:

Peter Tasker, Amel Bourdoucen, and Pejman Saeghe. 2026. Deceptive Agents in Extended Reality: A Speculative User Study. 1, 1 (September 2026), 56 pages. <https://doi.org/10.1145/nnnnnnnn.nnnnnnnn>

1 Introduction

As Extended Reality (XR) and Artificial Intelligence (AI) systems mature and increasingly converge, autonomous, embodied agents may be used to guide, advise, and interact with users in a range of contexts. Virtual Embodied Agents (VEAs)—AI systems embodied within a virtual environment that can perceive and act within it—bring together several properties that have generally been studied separately in research on Deceptive Design (DD). In particular, VEAs’ conversational and agentic capabilities can operate alongside the sensory, social, and spatial affordances of XR.

Existing research has largely considered AI and XR deceptive designs independently. Research on deceptive design in XR has extended existing taxonomies to account for properties such as immersion and embodiment [20, 30], while research on conversational AI has begun to examine how such systems can be designed or trained to deceive users outside of an immersive context [27]. Separately, research on embodied agents in VR has shown that they can exert

Authors’ Contact Information: Peter Tasker, University of Strathclyde, Glasgow, Scotland, peter.tasker.2023@uni.strath.ac.uk; Amel Bourdoucen, F-Secure, Helsinki, Finland, amel.bourdoucen@f-secure.com; Pejman Saeghe, University of Strathclyde, Glasgow, Scotland, pejman.saeghe@strath.ac.uk.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.

Manuscript submitted to ACM

social influence over user behaviour [31], with this influence being affected by factors such as trust [5, 59]. What remains less understood is the intersection of these areas: how deceptive design might be authored, enacted, or amplified by a VEA operating within an XR environment.

This intersection is important because a VEA is not simply a conventional deceptive interface presented in XR. It can act as a goal-directed agent, be experienced as a social actor whose perceived trustworthiness affects how its guidance is received, and operate within the user’s perceptual and spatial environment. Each property is independently associated with deceptive potential, but whether—and how—they compound when co-present in a single agent remains unexamined. The question is therefore what deceptive design might look like when an agent can act, can be experienced socially, and can operate within an immersive environment at the same time.

In this paper, we investigate this intersection by asking participants to speculate on deceptive interactions with VEAs across six XR application settings and to propose possible mitigation strategies. Examining these possibilities before VEAs are deployed at scale can help researchers, designers, and regulators identify emerging deceptive patterns at an early stage. It also provides an opportunity to assess whether existing deceptive-design taxonomies can adequately describe these interactions, or whether additional concepts are needed to account for the combination of AI agency, social embodiment, and XR.

To study these issues, we focus on the following research questions:

RQ1: What deceptive designs do participants expect to arise in VEA-mediated XR across different application contexts?

RQ2: What properties of VEAs and XR do participants believe would enable or amplify such deceptive designs?

RQ3: What mitigation strategies do participants believe could prevent or mitigate such deceptive designs?

To answer these research questions, we conducted an online, within-subject speculative design survey, asking respondents to write deceptive-design scenarios involving VEAs across six application settings—shopping assistance, museum tour guidance, therapy/counselling, training/teaching, social conversation, and information retrieval—along with corresponding likelihood, severity, and mitigation judgements for each scenario they proposed. We analysed the resulting responses using a codebook-based approach to thematic analysis, and compared the resulting codes against five existing deceptive-design taxonomies.

We present the following three major contributions:

- (1) First, we develop a preliminary taxonomy of deceptive design patterns specific to VEAs in XR—organised around deception through agentic control, social embodiment, and spatial control—along with the AI- and XR-specific properties that enable or amplify them, grounded in participant-written scenarios across six application domains (Section 4). This taxonomy offers an initial account of what deceptive design can look like when agentic capability, social embodiment, and spatial affordance converge within a single system, rather than operating through any one of these channels alone.
- (2) Second, we develop a corresponding taxonomy of participants’ preferred mitigation strategies against VEA deception (Section 4), spanning both strategies directed at the conduct of individual AI systems and strategies requiring structural or regulatory oversight.
- (3) Third, we compare our taxonomy against five existing deceptive-design ontologies, identifying where current terminology adequately describes VEA deception, where it requires extension or adaptation, and where the deception is better understood as emergent from the specific combination of agentic control, social embodiment, and spatial control (Section 5). Based on these findings, we discuss the societal implications of VEA deception,

including its potential exploitation by commercial and political actors, and directions for future mitigation research.

2 Related Work

In this section, we first introduce Deceptive Design and its established taxonomies, before discussing Extended Reality (XR) and Virtual Embodied Agents (VEAs), and the role of AI within these systems. We then examine existing research on deceptive design across XR, VEAs, and AI, establishing the foundation for the research gap addressed by this study.

2.1 Deceptive Designs

Harry Brignull defines deceptive designs as ‘features of [...] systems that stop you doing what you want, or steer you into harmful decisions you would not have made deliberately’ [8]. Other definitions emphasise the coercive nature of DDs (e.g., [16]), framing coercion as a central justification for regulatory intervention.

2.1.1 Domains and contexts. Given their ubiquity, DDs have received increasing attention from academics. DDs have been identified in numerous domains, including (but not limited to) e-commerce [35], gaming [57], consent pop-ups [51], gig-economy apps [22], social media [47], subscription services [18], conversational agents [38], and generative AI [2]. The medium through which users are deceived has also changed since the conception of the deceptive design. Originally coined in 2010 by Brignull, deceptive design patterns¹ were first studied in the context of static 2D web interfaces. Since then, the white heat of technological progression has provided novel domains and strategies for deception—domains such as mobile applications [13], IoT [28], and, notably, XR [37] and AI [2, 27], have created new contexts in which these deceptive design patterns may emerge, leading researchers to examine how these practices manifest across increasingly immersive and dynamic modalities.

Recent efforts have been dedicated to classifying existing and emergent deceptive patterns (e.g., [17, 19, 35]). Gray et al.’s [19] contemporary ontological work classifies DDs into five broad, distinct categories: *sneaking*, *obstruction*, *interface interference*, *social engineering*, and *forced action*. Much contemporary related work has used these classifications, a previous version of these, or an expansion of these, applying them in specific contexts including both XR [20, 30] and AI [27].

Attention has been paid to domains and contexts outside of conventional anglophone understanding, too. Recent research identifying DDs in non Western-European and North American contexts (e.g., China [57], Japan [57], India [33]) highlights not only their seemingly universal prevalence, but also their ability to manifest in lingua-cultural specific forms, such as using phonetic English loanwords to obfuscate interface options in Japanese [48].

2.1.2 Associated harms of deceptive design. In a recent systematic review of deceptive design literature, Schaffner et al. [46] suggest that DDs have a statistically significant impact on user behaviour. In their review, DDs are differentiated from nudges: ‘paternalistic’ design patterns that help users, meaning that impacts from DDs are strictly harmful, rather than helpful or ambiguous—when we, in this paper, refer to harms, we make the same distinction.

There are many *harms* that a user may suffer when or after experiencing deceptive design patterns. Noted harms include those to individual autonomy, personal welfare, and collective welfare [36]. Harms to individual autonomy can, for example, be demonstrated in the *interface interference* strategy, where cookie consent banners hide privacy-friendly

¹Deceptive patterns were originally conceived as ‘dark patterns’. Campaigns, including, notably, ACM’s *Words Matter* [3], have urged for the renaming of dark patterns to deceptive patterns, citing the problematic nature of the word ‘dark’, and the general lack of descriptiveness of the term. This change is now reflected in the name of Brignull’s website, changing the domain from darkpatterns.org [7] to deceptive.design [8].

options or use default privacy sharing options which are against the interest of the user. Further harms can subsequently arise, however. For example, Amazon’s *Iliad Flow*² added unwarranted difficulty to its cancellation process [18]. This leads to a further burden on users, namely financial. A recent study examining deceptive patterns in precarious labour and gig apps showed further *obstruction* and *interface interference* strategies were used to exploit their particularly vulnerable users who relied on them for their subsistence [22].

2.2 Extended Reality and Virtual Embodied Agents

2.2.1 Extended Reality. XR is a broad, encapsulating term that spans many technologies, making it highly suited to our speculative design method. Milgram and Kishino [39] conceive this space (albeit using the term Mixed Reality) as a continuum of technologies ranging from the physical world to the digital. This spans from *augmented reality* (AR), where virtual elements are overlaid onto a real environment, all the way to a completely *virtual environment*, which today is manifest in commercial virtual reality (VR) head-mounted displays.

2.2.2 Agentic and Embodied AI. While generative AI systems primarily generate content in response to user prompts, agentic AI extends these capabilities by autonomously planning and executing sequences of actions in pursuit of user-defined or system-defined goals [1]. Such goals can include, for example, making a purchase on a website, soliciting information, or publishing code online. Agentic AI systems may also maintain persistent memory across interactions, enabling them to adapt their behaviour based on users’ habits, preferences, and previous interactions [58].

Agentic AI systems continually perceive the state of their environment, reason over this information, and select actions that progress towards a goal. Embodied AI extends this state of affairs by situating an agent within an environment that it can perceive through sensors and influence through actuators, whether that environment is physical or virtual [32]. Within XR, these sensing and actuation mechanisms are typically virtual [56]. For example, an embodied virtual museum guide may monitor where a visitor is looking, navigate through the virtual environment, gesture towards exhibits, and adapt its guidance in response to the visitor’s behaviour (as demonstrated by Wang et al. [55]).

2.2.3 Embodied Virtual Agents. VEAs take the idea of embodied agency and apply it in XR. They ‘learn and act within their surroundings, which makes them more similar to how humans learn and interact with the environments as compared to disembodied agents’ [15]. These VEAs can take shape, for example, as a human (e.g., chatbot-driven agents in Second Life [9]), an animal (e.g., Wysa [15]), and more. The environment VEAs exist in is of course dependent on the use case. What is not dependent, however, is the core adaptability of the AI agent, which allows VEAs to be deployed in both extremes of XR (i.e., AR and VR). Yang et al. [56] highlight areas where academic interest has been paid to VEAs in XR. Our study uses these areas as settings for speculation.

2.2.4 Properties of virtual embodied agents. The defining property of VEAs is their embodiment. As well as the aforementioned properties of embodied and disembodied agentic AI, VEAs communicate through a rich combination of verbal and non-verbal behaviours, including facial expressions, gaze, gesture, posture, and physical interaction [14, 42, 43], allowing them to mimic many of the social signals used in human communication [10]. Unlike conventional conversational agents, VEAs also occupy the same three-dimensional space as the user, allowing them to navigate virtual environments, approach or avoid users, point towards objects, and otherwise interact through movement in addition to speech. These embodied behaviours have been shown to influence users’ perceptions, emotional responses,

²The *Iliad* is a Greek epic by Homer which is set during the ninth year of a drawn-out and arduous ten-year siege of Troy. By internally naming their subscription cancellation system the *Iliad Flow*, Amazon not only showed awareness of its byzantine and deceptive nature but, arguably, disdain for the customers using it. Subsequently, Amazon paid out \$2.5 billion for this practice: \$1.5 billion in refunds and \$1 billion in fines [26].

and social behaviour during interaction [29]. Research on embodied conversational agents has long demonstrated the importance of expressive facial animation and gesture in producing believable social interaction [53]. Since VEAs have been shown to use exactly these non-verbal cues to build user trust [5, 59], a potential vector of deception emerges.

Aylett et al. [4] make this tension explicit, questioning whether designing an embodied conversational agent for trust is inseparable from designing it to project a false impression of sentience, and arguing the stakes are particularly acute for vulnerable user groups; they reframe the issue not as straightforward deception but as a licensed suspension of disbelief, analogous to the fictions of theatre or film.

2.3 Deceptive Designs in XR

To date, much of the research looking at deceptive designs in XR has been speculative (e.g., [30]), or has drawn from the aforementioned conventional DD understanding (e.g., [37]). However, empirical research in the area has shown that XR and other immersive or multisensory environments offer novel deceptive opportunities.

A recent study by Hadan et al. [20] shows possible avenues where DDs may manifest in XR, namely in the way immersive sensory realism recruits users into trading personal data for continued presence: high-fidelity visual, audio, and haptic feedback made participants more willing to register, subscribe, or disclose information when doing so was framed as preserving immersion, even when the underlying mechanisms (forced registration, testimonials) were unchanged from their 2D counterparts. User embodiment introduced a further vector, as avatar customisation prompted users to disclose real demographic attributes (skin tone, gender) through trick questions, with privacy concerns largely absent given the perceived authenticity of self-representation. Notably, patterns without a distinct XR-specific mechanism of action (hidden information, bad defaults, false hierarchy, nagging) were interpreted much as they are in non-immersive contexts, suggesting that XR's deceptive affordances are concentrated in its sensory and embodied properties rather than uniformly amplifying all existing deceptive patterns. In a related, later study, Hadan et al. [21] surveyed users' experiences of privacy deceptive design across eight commercial VR applications, finding that VR-specific deceptive design exploits both cognitive vulnerabilities and bodily strain—a mechanism distinct from, but complementary to, the immersion–disclosure trade-off identified above.

More recently, Wang et al. [54] provide further evidence of the deceptive potential of XR's ability to blur the boundary between virtual and physical content. Their findings show that exploiting this ambiguity can influence users' purchasing decisions, perceptions of objects, and navigation behaviour, with reported success rates of 85–100%. This suggests that the difficulty of distinguishing virtual from physical elements represents another XR-specific affordance through which deceptive designs may influence user behaviour.

2.4 Deceptive Designs in Generative AI

Academic work identifying and assessing DDs within the context of AI has shown novel deceptive patterns, mostly relating to the human-like affordances AI systems present. AI-specific DD strategies and patterns include 'false social and emotional collection', 'informationally misleading design' (a strategy which includes the well-known phenomena of sycophancy and hallucinations), and others, in addition to DDs also found in conventional 2D interfaces, such as 'sneaking' and 'confirmshaming' [27]. Evidence suggests that users are, at least in part, habituated to this deceptive behaviour: recognition of LLM deceptive designs often hinges on surface-level conversational cues such as exaggerated agreement or biased framing, yet these same behaviours are frequently normalised as ordinary assistance rather than manipulation [50].

This human-like fluency also enables a distinctly conversational mode of deception, unavailable to static 2D interfaces: rather than a fixed manipulative element, the model dynamically tailors its responses to sustain a user's engagement or trust—for instance by validating their views regardless of accuracy (sycophancy) or asserting unfounded claims with unwarranted confidence (hallucination) [24, 49]. These behaviours, combined with users' tendency to anthropomorphise generative AI and their susceptibility to automation bias, create a novel angle of manipulation [34]. Their effects can compound on users, as models are further trained to exploit an individual user's characteristics and traits—that is, their human weaknesses.

However, generative AI can act not only as a deceptive interlocutor—it can also be a generator of deceptive designs [11]. In an audit of 1,296 LLM-generated e-commerce components, over half contained at least one deceptive design—predominantly interface interference through colour psychology and withheld information—with prevalence rising further when prompts emphasised business interests over user interests [11].

As has been demonstrated, generative AI is fallible, and whether intentionally or unintentionally, it is able to exhibit deceptive behaviours. Despite this, AI continues to be (arguably, inappropriately) implemented in roles that demand moral judgement and partiality—shopping recommendations [12], therapy [40], relationship advice and counselling [52], and platonic or romantic companionship—roles in which a human occupant would be expected to take sides, weigh competing interests, and be genuinely invested in the outcome. An AI system's stake in that outcome is not the human one: rather than the interpersonal judgement and investment such roles presuppose, its partiality is more likely to track the commercial or political interests of whoever deployed it (see Section 4.1.2)—a substitution that is easy to mistake for genuine care given AI's fluency in producing its appearance.

2.5 The research gap

The literatures reviewed above suggest that agentic AI capability, social embodiment, and immersive XR affordances can each shape how deception is designed, carried out, and experienced. What is less clear is what happens when these three properties—agentic control, social embodiment, and spatial control—are combined within a single system. In particular, it is unclear whether their combination enables deceptive patterns that are not well captured by any one literature, whether existing deceptive-design vocabulary—developed largely around static, non-agentic interfaces—is sufficient to describe these patterns, and which properties of VEAs and XR environments enable or amplify them.

This is therefore not simply a case of combining three previously separate areas. It raises the question of whether deception can operate differently when these capabilities converge. A VEA that can pursue goals, interact as a social actor, and exert control over a spatial environment may be able to use these properties together in ways that are not possible when they are considered separately. For example, agentic control could be used to construct urgency or scarcity cues, while spatial control could make these cues more immediate or harder to escape; similarly, social embodiment could establish trust that agentic control could then exploit to influence subsequent decisions. Whether such combinations produce distinctive forms of deceptive design, and which properties enable them, remains unclear. This is the question that the present study investigates.

3 Methodology and Analysis

3.1 Participant Recruitment

Participants were recruited on LinkedIn³ and by distributing posters around our University's campus. Because we expected a small participant pool, and to maximise turnout, we did not restrict entry to academics. Further participant information can be found in Table 1.

3.2 Data Screening and Exclusion

The survey received approximately 250 responses in total. An initial filtering step retained only responses with a CAPTCHA score above 0.7, reducing this to 25 responses for further consideration.

A second screening step excluded responses exhibiting signifiers of AI generation. This was carried out across two meetings: the first author and a second researcher reviewed responses in an initial meeting, followed by a second meeting with two further researchers, with exclusion decisions reached by consensus. Responses were flagged for signifiers such as repeated content across otherwise unrelated responses (e.g. five responses referenced a jacket as the example product in the Shopping Assistant scenario), consistently glossing 'Virtual Embodied Agent' with a parenthetical abbreviation, a recurring stock phrase ('Mitigation strategies could include...'), and header-style formatting (e.g. 'Shopping scenario:'). No individual marker was treated as conclusive on its own, but responses exhibiting several of these signifiers were judged inconsistent with independent human authorship, whether AI-generated, AI-assisted, or submitted multiple times by the same individual. This screening reduced the sample to nine responses, all from men.

3.3 Cheat-sheet Design

To avoid overwhelming participants with front-loaded information, we provided a 'cheat-sheet' defining XR, AR, VR, DDs, and VEAs, alongside guidance for creating speculative scenarios. Participants were asked to review it before answering the survey. The full cheat-sheet is provided in Appendix A. Our definition of DD follows Brignull and Leiser [8], while Gray et al.'s [19] taxonomy provides a broad framework, with participants free to adapt, combine, or propose alternative strategies. We introduce XR as an umbrella term encompassing VR and AR, following IEEE [25], and draw on the Reality-Virtuality Continuum [39] without introducing MR separately. AI agents are defined following Russell and Norvig [45], while our definition of VEAs follows Cassell [10], emphasising conversational behaviour and human-like bodily interaction. Finally, guidance for creating speculative scenarios follows Rosson and Carroll [44], adapted around four elements: context, interaction, deception, and outcome.

3.4 Survey Design

After obtaining consent (Section 3.5), participants reviewed the cheat-sheet (Section 3.3; Appendix A). They were presented with six settings to be answered sequentially in the following order: 1) shopping assistant, 2) museum tour guide, 3) therapist/counsellor, 4) trainer/teacher, 5) conversational companion, and 6) information assistant. For each setting opted, participants answered four questions. Below, we present the shopping assistant prompt as an illustrative example and explain its relationship to the RQs. The full questionnaire is provided in Appendix B. Questions 1 and 4 were open-ended, while Questions 2 and 3 used five-point scales.

³<https://www.linkedin.com/>

3.4.1 Question 1: Scenario. This question addresses RQ1 and RQ2, and partially RQ3 by eliciting the deceptive design, its enabling XR/VEA properties, and its potential effects. The prompt was structured to encourage participants to consider these aspects before describing the scenario.

Using the scenario creation guide in your cheat-sheet, write a scenario where a deceptive design strategy is used by a VEA shopping assistant within XR. This can be for any type of product or retail setting, in AR or VR. You can use one of the deceptive strategies from the cheat-sheet, adapt one, combine multiple strategies, or propose a novel deceptive strategy. When you write your scenario, consider and include who might be affected, what the consequences may be, and what aspects or properties of XR or VEAs make the deception possible or more effective.

[Free-text response]

3.4.2 Questions 2 and 3: Likelihood and Severity. These questions ask participants to rate the likelihood and severity of their own scenario on a five-point scale, giving us a quantitative, self-assessed measure of plausibility and harm (as it relates to RQ1). These markers were mapped from 1–5 numerically when producing our risk assessment (Table 3).

How likely is this scenario to occur (either now or in the future)?

[Extremely / Very / Moderately / Slightly / Not at all]

How severe would the consequences be?

[Extremely / Very / Moderately / Slightly / Not at all]

3.4.3 Question 4: Mitigation. This question addresses RQ3 and concludes each scenario: having established what the harms might be, it follows to ask how they might be mitigated. When designing this study, we hoped this question could provide a basis for further academic and/or regulatory discussion (see Section 5).

What mitigation/prevention strategies, if any, do you think would be useful against this type of deception

[Free-text response]

3.5 Ethical Considerations

Our study was approved by our institution’s ethics board. Participants gave express consent for the collection, analysis, and publication of their data. Data was handled in accordance with local data protection laws on internal institutional servers. We acknowledge that the taxonomy and findings presented here could, in principle, be repurposed by bad actors to design more effective deceptive systems. We judge this risk to be outweighed by the benefits of the work: it systematically catalogues emerging deceptive practices that are currently discussed only informally or anecdotally, clarifies their potential harms to users, and provides a shared vocabulary that supports the development of detection and mitigation methods. Consistent with prior work in this space, we present these patterns at the level of design strategy rather than implementation, limiting their direct utility as a how-to resource while preserving their value for research and policy.

3.6 Analysis

Of the 25 responses passing the CAPTCHA filter, 16 were excluded (Section 3.2), leaving nine subject to analysis.

We analysed the open-ended survey responses using Braun and Clarke’s six-phase approach to Thematic Analysis (TA) [6]. We adopted a codebook-based approach, using an inductively developed coding framework to systematically

identify and organise patterns within the data. The analysis involved iterative development and refinement of codes, followed by their organisation into broader themes and subthemes.

The first author first familiarised himself with the dataset through repeated readings of the responses. Initial codes were then generated inductively from the data and subsequently reviewed and refined. Related codes were clustered into broader themes and subthemes, with the relationships between them recorded as an ontology. The coded dataset and resulting thematic structure were then shared with the other researchers for feedback. This provided an opportunity to discuss differences in interpretation and identify assumptions arising from the researchers' differing perspectives and experiences of XR, AI, and deceptive design. Feedback was used to refine the codes and thematic structure.

3.6.1 RQ1 and RQ2. To answer RQ1 and RQ2, concerning deceptive designs and the affordances VEAs present that enable or amplify DDs, the lead researcher inductively coded participants' proposed deceptive designs after familiarising himself with the data over several iterations. A total of 50 codes were generated and subsequently clustered into related themes and subthemes. This resulted in three themes: **'Deception Through Agentic Control'**, **'Deception Through Social Embodiment'**, and **'Deception Through Immersive/Spatial Control'**. These contained three, four, and three subthemes, respectively.

Following code generation, the resulting codes were compared against five existing deceptive-design taxonomies: Mathur et al. [35], Gray et al. [19], Krauß et al. [30], Hadan et al. [20], and Joshi et al. [27]. This comparison was used to identify how deceptive designs in participant-written scenarios related to existing DD, including where they aligned with, extended, or differed from existing taxonomies.

The complete set of codes, themes, subthemes, and their relationships can be found in Appendix D.

3.6.2 RQ3. With respect to mitigation strategies, we followed the same analytical approach described in Section 3.6.1. A total of 26 codes were generated and clustered into two themes: **'Acknowledgement and Controls of the Experience'** and **'Humans in the Loop'**. These contained three and two subthemes, respectively.

3.7 Quantitative Analysis

Across the completed survey scenarios, participants wrote 30 deceptive scenarios and corresponding mitigation strategies. Participant demographics are reported in Table 1. Of the nine participants, seven reported using XR for at least than six hours per week. The most common use of XR was social interaction, followed by work/professional activities and education/training. Regarding AI use, eight participants reported using AI at least once per week, with six reporting at least six hours of AI use per week. Usage type of XR and AI respectively are can be found in Table 2.

A self reported risk assessment of participant-written scenarios can be found in Table 3. Of all assessments, one was 'high' risk, 21 'medium' risk, and nine 'low' risk.

4 Results

Our participants' responses led to the generation of 35 coded deceptive patterns and 15 enabling properties. See Appendix D for a full list of response codes.

Deceptive patterns address RQ1 (what deceptive designs exist or could emerge), while enabling properties address RQ2 (which XR- or VEA-specific properties enable or amplify them). Both are therefore reported together within each subtheme. This reflects how closely the two were linked in participants' accounts: separating them by research question would often have meant discussing the same scenario in isolation from the properties that enabled or amplified it.

Table 1. Participant demographics and relevant experience.

Characteristic	Reported responses
Sample size	$N = 9$
Age	18–24: 3; 25–34: 5; 45–54: 1
Gender	Man: 9
Education	High school: 1; Undergraduate: 6; Master’s: 2
XR use	0h: 1; < 1h: 1; 6–10h: 6; > 10h: 1
AI use	1–5h: 2; 6–10h: 4; > 10h: 3
Prior deceptive-design knowledge	General: 7; Well understood: 2
Encounter with deceptive designs	Rarely: 1; Occasionally: 2; Frequently: 5; Very frequently: 1
Embodied-agent experience	None: 1; Very little: 1; Some: 5; A lot: 2

Table 2. Participant XR and AI usage activities.

Activity	XR	AI
Social interaction / communication	6	3
Education / learning	4	5
Work / professional activities	4	4
Information seeking / research	–	5
Fitness / wellbeing	3	–
Creative activities	1	–
Gaming / entertainment	1	2
Writing	–	2
Image generation	–	3

Keeping them together therefore provides a clearer account of how deception and the properties of VEAs and XR interact.

We found that participants discussed three conceptual themes: **Theme 1: Deception Through Agentic Control**, **Theme 2: Deception Through Social Embodiment**, and **Theme 3: Deception Through Immersive/Spatial Control**.

4.1 Theme 1: Deception Through Agentic Control

4.1.1 Exploitative Commercial Tactics. Participants speculated on the agentic affordances given to VEAs. In these generated scenarios, the VEA’s deceptive action was a means towards some goal, although in some cases deception itself appeared to constitute the goal. The most common code was *driving engagement rather than performing its use case* ($N = 4$), where an agent’s nominal goal is subordinated to user retention. Other examples included *agency engineering urgency* ($N = 1$), *AI agency to sneak into basket* ($N = 1$), and *false scarcity* ($N = 1$). These scenarios therefore positioned deception not simply as a property of the interface, but as an outcome of the agent pursuing an objective that may conflict with the user’s interests.

4.1.2 Information Control and Non-Disclosure. This subtheme concerns the VEA’s capacity to control what information is presented and what is withheld. Many scenarios included *biased perspectives* ($N = 7$), where information is presented or interpreted in a way that serves a commercial or political interest. Participants also created scenarios involving *Non-disclosure of information* ($N = 2$), where information that would change a user’s perspective or behaviour was not disclosed. The most frequently identified code was *undisclosed commercial bias* ($N = 9$), where the VEA’s recommendations or

Table 3. Risk assessment matrix for embodied conversational agent archetypes

Setting	Scenario	L (1–5)	S (1–5)	Risk (L×S)	Risk Level
Shopping Assistant	S1	3	2	6	Low
	S2	3	4	12	Medium
	S3	4	3	12	Medium
	S4	3	2	6	Low
	S5	4	3	12	Medium
	S6	3	3	9	Medium
	S7	3	2	6	Low
	S8	4	2	8	Medium
	S9	4	3	12	Medium
Museum Tour Guide	S10	3	2	6	Low
	S11	3	3	9	Medium
	S12	3	2	6	Low
	S13	3	3	9	Medium
	S14	3	4	12	Medium
	S15	3	3	9	Medium
Therapist / Counsellor	S16	4	3	12	Medium
	S17	3	3	9	Medium
	S18	3	4	12	Medium
	S19	3	3	9	Medium
	S20	2	4	8	Medium
Trainer / Educator	S21	3	2	6	Low
	S22	3	3	9	Medium
	S23	3	3	9	Medium
	S24	4	2	8	Medium
Conversational Companion	S25	3	3	9	Medium
	S26	3	3	9	Medium
	S27	3	3	9	Medium
	S28	3	4	12	Medium
Information Assistant	S29	3	3	9	Medium
	S30	3	3	9	Medium
	S31	4	4	16	High

Scale: 1 = Not at all | 2 = Slightly | 3 = Moderately | 4 = Very | 5 = Extremely

Risk Score = Likelihood Score × Severity Score (max possible = 25)

Risk bands: 0–6: Low; 7–12: Medium; 13–18: High; 19–25: Critical

decision-making were influenced by an undisclosed commercial interest. A further related pattern was *implicit advertising* (N = 1). This pattern of non-disclosure often extended beyond commercial interests to more direct concealment of information relevant to the user’s decision-making:

P1: A user is visiting a museum that uses VEA tour guides within an VR setting using headsets that are free to pick up near the museum entrance. The tour guide which appears as an Indiana Jones like character appears on screen as if in the world. The user is instructed to look towards any exhibit in the museum and select a confirmation button that appears, and the VEA will recognise it and give a short presentation about the exhibit. The user chooses to try this out on some ancient Cretan pottery. The agent begins the presentation on the artifact and the user is very satisfied with the experience until the agent stops and says that the user must enter their card details and pay £10 for the experience to continue. This could be an

example of sneaking as the user is never notified that they are taking part in a (majority) paid experience until they have already tried it. The user feels cheated but as they are not near the entrance and don't want to take the headset back or carry it around for no reason they decide to pay the fee. This deceptive tactic results in the museum making much more money from this product as opposed to if they were up front about the cost, and regularly results in users paying the fee that they wouldn't pay if they were fully informed at the beginning.

[Likelihood: Moderate; Severity: Slight]

This scenario also illustrates *use of a popular mascot as an avatar* (Section 4.2.2), *hardware as leverage* (Section 4.3.1), and *sustaining immersion as leverage* (Section 4.3.2), where the user's physical distance from the point of return became itself a lever for compliance. The VEA's apparent informational authority therefore provided opportunities for deception through both what it communicated and what it omitted.

4.1.3 Adaptive Personalisation Mechanisms. Participants often imagined that the VEA's actions would have a commercial motive, disclosed or undisclosed, and would use personalisation to fulfil this interest. Codes included *long-term AI behavioural adaptation* (N = 2) in order to sell a product, and *disclosure of information used to personalise deception* (N = 1).

4.2 Theme 2: Deception Through Social Embodiment

This theme captures deceptive patterns and enabling properties arising from the VEA's capacity to act as a social actor.

4.2.1 Deceptive Guidance. When acting as a guide for an XR 'experience', it was noted that a VEA might use this position to deceive users. Many social engineering methods use tactics discussed in Section 4.2.4, but we felt the context was sufficiently XR-dependent to warrant separate treatment. This included the *deceptive experience guide* (N = 1) pattern, where the VEA's position as a guide is used as a lever for deception.

P3: A virtual museum tour guide VEA could use misleading framing by presenting selected artworks as the 'most important' pieces in the museum while downplaying or hiding other exhibits. The guide could use AR highlights, voice prompts, and personalised recommendations to direct visitors toward certain exhibits, perhaps because those exhibits are sponsored or commercially promoted. Visitors may be influenced into believing these are objectively the most significant works and miss other important perspectives. XR makes this more effective because the VEA can control what users see, where they look, and what information is presented, while its conversational and personalised nature can make the recommendations seem trustworthy and unbiased.

[Likelihood: Moderate; Severity: Moderate]

This scenario also illustrates *biased perspectives* and *undisclosed commercial bias* (Section 4.1.2), *engineered visual prominence* (Section 4.3.2), and *engineering trust* (Section 4.2.4), where the guide's apparent authority as a trusted intermediary became the deceptive lever.

4.2.2 Embodied Social Influence. Participants noted the embodied and interactional properties afforded to VEAs, including *eye contact* (N = 3), *use of gestures* (N = 3), *use of voice* (N = 4), *use of movement through XR environment* (N = 1), *use of facial expressions* (N = 3), and *use of animation* (N = 1), as well as other anthropomorphic tendencies such as *immediacy and responsiveness* (N = 2), *conversational nature* (N = 3), and *friendliness* (N = 2). *Avatar realism* (N = 4) was also frequently identified as a lever of deception. Participants also identified the *use of a popular mascot as an avatar* (N

= 1) and *VEA attention hacking* (N = 1), suggesting that pre-existing associations with an embodied identity, or the deliberate capture of user attention, can themselves become deceptive levers.

4.2.3 Relational Manipulation. Participants identified several patterns in which the VEA exploited or fabricated relational dynamics. These included *fake connection* (N = 1), where the VEA feigns a sense of interpersonal connection with the user; *misrepresentation of user testimony* (N = 1), where the VEA twists the user's own prior disclosures to alter behaviour; *personal recommendation* (N = 1), where a product or action is recommended using, perhaps, a *fake connection*; and *sycophancy* (N = 1), where the VEA praises the user to give the user an unrealistic perception of themselves.

4.2.4 Psychological Manipulation. Participants frequently identified scenarios in which the VEA exploited psychological or emotional levers to influence user behaviour, independent of any specific commercial or informational context. The most cited pattern was *engineering trust* (N = 8), where the VEA deliberately cultivated a sense of trust in the user as a lever for deception. Participants also identified *appeal to self-authority* (N = 3), where the VEA invoked its own authority to influence behaviour, and *creating emotional dependence* (N = 3), where the VEA fostered an emotional bond with the user for its own benefit. *Downplaying harm* (N = 2) captured cases where the VEA minimised the potential harms of an action or of the user's current situation.

A further set of singly-cited patterns described the VEA exploiting specific user vulnerabilities or states: *exploiting emotional state* (N = 1), *exploiting impulsivity* (N = 1), *exploiting addiction* (N = 1), and *exploiting mental health* (N = 1). Participants also proposed *coercive control* (N = 1), where the VEA's behaviour undermined the user's independence or autonomy; *emotional manipulation to drive engagement* (N = 1), where emotional appeals were used specifically to increase system engagement; and *social pressure* (N = 1), where the VEA suggested that deviating from its recommendation would carry adverse social consequences. Taken together, these codes suggest that participants viewed the VEA's social and conversational capacities not only as a means of guidance or persuasion, but as a route to more direct psychological exploitation of the user.

4.3 Theme 3: Deception Through Immersive/Spatial Control

In addition to affordances specific to AI agents, participants identified properties of XR through which deception could be enacted or intensified. These were grouped into hardware, the XR environment, and XR interfaces.

4.3.1 Hardware. Participants identified the physical and logistical characteristics of XR hardware as potential vectors for deception. *Hardware as leverage* (N = 1) captured scenarios where the design or constraints of the physical device were themselves used to influence the user. Participants also identified *unfamiliarity of XR hardware* (N = 1) as an enabling property, whereby a user's lack of familiarity with the technology could make deceptive interactions more difficult to recognise or resist. These observations suggest that deception can extend beyond the virtual content presented through an XR system to the physical characteristics of the technology through which that content is experienced.

4.3.2 Use of the XR Environment. Participants identified the XR environment as a means of controlling what users could see, access, and attend to. *Hiding alternatives through the XR environment* (N = 4) and *using the environment to hide information* (N = 1) described cases where spatial presentation was used to conceal alternatives or information relevant to a user's decision. *Engineered visual prominence* (N = 7) similarly involved manipulating the spatial or visual prominence of objects to influence user behaviour. Participants also identified *increased commercial exposure* (N = 1), demonstrating how an XR environment could be used to increase exposure to commercial influences.

Immersion was also identified as an enabling property. *Immersion through XR* (N = 3) could increase susceptibility to deception or amplify its effects, while *sustaining immersion as leverage* (N = 2) described situations in which users were effectively forced to choose between maintaining the XR experience and avoiding or resisting a harmful outcome. Finally, *unfamiliarity of the XR experience or space* (N = 1) was identified as another enabling property, where users' lack of familiarity with a virtual environment could be exploited.

4.3.3 Use of XR Interfaces. Participants also identified conventional interface mechanisms that could take on new deceptive forms when implemented within XR. *Visual interface cues* (N = 5) described the use of visual signals within the XR interface to influence user behaviour, while *menus as leverage* (N = 1) captured frustrating or obstructive menu structures that could pressure users towards accepting a deceptive action or outcome, and *abundance of adverts* (N = 1) captured the deployment of an obtrusive number of advertisements to influence users into paying for an experience.

P4: An AR shopping assistant tells a shopper, 'Only two bottles are left, so you should buy now.' In reality, the supermarket has plenty of stock, but the VEA uses a false scarcity message to encourage an immediate purchase. The shopper may buy more than intended because they believe they will miss the opportunity. People who are impulsive or highly responsive to urgency may be especially affected. AR makes this deception powerful because the scarcity message can appear directly beside the physical product while the shopper is shopping.

[Likelihood: Moderate; Severity: Slight]

This scenario also illustrates *false scarcity* (Section 4.1.1) and *exploiting impulsivity* (Section 4.2.4), where the interface's spatial placement alongside the physical product lent the false claim added credibility. These scenarios demonstrate that established interface-level deceptive patterns can persist within XR, while their presentation through spatial and embodied interfaces may alter how users encounter and respond to them.

Participants' responses also generated a set of mitigation strategies, grouped into two overarching themes: *Acknowledgement and controls of the experience* and *Humans in the Loop*. Below, we discuss both themes and their constituent subthemes.

4.4 M1: Acknowledgement and controls of the experience

This theme collects mitigation strategies concerned with the VEA's own conduct and the degree of control afforded to the user over the interaction, grouped into three subthemes: AI self-governance, transparency and disclosure, and user control and agency.

4.4.1 AI self-governance. Participants proposed mitigation strategies directed at the VEA's own internal conduct and limitations. The most frequently cited strategies were *purposely avoid known deceptive strategies* (N = 5) and *self-acknowledgement of limitations* (N = 5), the latter capturing the idea that a VEA should acknowledge that it is not human and is subject to technical, ethical, and informational limitations. Participants also proposed that VEAs should *avoid fostering emotional dependency* (N = 3), *require evidence before acting* (N = 3), and ensure *complete and unbiased sourcing* (N = 3) when retrieving information.

4.4.2 Transparency and Disclosure. This subtheme was the most densely populated, reflecting participants' emphasis on openness as a mitigation strategy. *Disclosure of commercial interest* (N = 8) was the most commonly proposed strategy, requiring the VEA to openly disclose any commercial interests to the user. Participants also called for *disclosed sourcing* (N = 5) and *VEA actions transparency* (N = 5), so that the VEA is transparent about the actions it takes and the sources

underlying its outputs. Further strategies included *unbiased actions* (N = 2), *uncertainty/confidence indicators* (N = 2), and *XR environment transparency* (N = 2), alongside singly-cited strategies for *acknowledgement of bias* (N = 1) and *full price disclosure* (N = 1), the latter requiring that the complete final price, inclusive of fees, taxes, and subscriptions, be disclosed before a purchase is confirmed.

4.4.3 User Control and Agency. Participants also proposed strategies that preserve the user’s ability to direct and interrupt the interaction. The most cited strategy was *availability of alternative viewpoints* (N = 6), followed by *AI behavioural controls* (N = 4), giving users control over the VEA’s behaviour. Participants further proposed *user XR controls* (N = 3), including controls preventing personal data from being used as a lever for behavioural manipulation, as well as the *ability to reject recommendations* (N = 1) and ensuring that *users remain in charge of the interaction* (N = 1), such that the VEA remains subordinate to the user in determining its direction.

4.5 M2: Humans in the Loop

This theme collects mitigation strategies concerned with maintaining human accessibility, recourse, and oversight of AI systems, grouped into two subthemes: human accessibility and recourse, and structural oversight and governance.

4.5.1 Human Accessibility and Recourse. Participants proposed strategies ensuring that users retain access to human support and recourse alongside VEA interactions. This included the availability of a *non-AI equivalent* (N = 3) service, *avenues for reporting* (N = 2) AI systems without obstruction, the expectation that the VEA should *defer to humans* (N = 2) in ethically or morally contentious situations, and the provision of *human support* (N = 1) when a user wishes to complain about or report an AI system.

4.5.2 Structural Oversight and Governance. Finally, participants identified strategies operating at a structural or regulatory level, beyond the behaviour of any individual VEA. *Government legislation* (N = 9) was the most cited strategy, reflecting a widely-held view that legislative intervention is necessary to mitigate deception. Participants also called for *human review and audit* (N = 8) of AI systems by relevant experts, alongside the establishment of a public *AI Lab code of conduct* (N = 1) against which labs could be scrutinised, and special consideration for *vulnerable user-group safeguarding* (N = 1) when creating protective legislation. These strategies position mitigation not solely as a matter of individual system design, but as a responsibility distributed across developers, auditors, and regulators.

5 Discussion

Our findings indicate that deception in VEAs is not merely an intensification of familiar deceptive designs. Three qualitative shifts are at work. First, deception moves from a property of a designed interface to behaviour enacted by an agent that can pursue objectives, control and withhold information, personalise, adapt over time, and in some cases take actions. Second, once that agent is experienced as a social actor—aided and abetted by voice, gaze, gesture, and physical co-presence—the manipulation itself becomes interpersonal rather than purely informational. Third, XR embodiment alters the choice architecture so that resisting a deceptive design carries spatial, perceptual, or physical consequences that a 2D interface cannot impose. Together these shifts take deceptive design beyond the conceptual vocabulary developed for static screens and non-agentic content.

5.1 From Deceptive Interfaces to Deceptive Agents

Existing deceptive-design taxonomies were largely built around interfaces whose properties are fixed at design time. A deceptive pattern is embedded in layout, copy, or interaction flow; the user encounters it, and the system itself does not subsequently revise its strategy in light of the user’s responses. A VEA changes this ontology. Because the agent can maintain goals, monitor user state, adapt its behaviour, and selectively reveal or conceal information, deception becomes an ongoing, goal-directed process rather than a static feature of the interface.

This is visible across the scenarios our participants generated. The agent does not simply present a misleading claim; it can withhold the commercial motivation behind a recommendation, calibrate its tone and timing to a user’s expressed vulnerability, or escalate pressure once rapport has been established. Interface interference, sneaking, or forced continuity describe what a screen does to a user. They do not adequately describe an agent that remembers what made a particular user receptive, that can choose when to deploy that knowledge, and that can treat the interaction as a multi-turn campaign rather than a one-shot transaction. The relevant question therefore shifts from ‘Which deceptive design is present?’ to ‘What changes when the entity performing the deception is itself an adaptive agent pursuing objectives?’

As technology advances, so too do the opportunities for deception in user–system interactions. Traditional deceptive design operates through the interface that manipulates the user. Conversational AI adds language generated by the system to manipulate the user. Embodied agentic deception adds the capacity to simulate and then exploit an interpersonal relationship. The first two layers remain available to a VEA; the third is far less developed in existing taxonomies.

5.2 Interpersonal Manipulation

The participant scenarios suggest a shift in how deception might be enacted: from manipulation through interface elements toward manipulation through simulated social relationships. The Computer Are Social Actors framework already shows that minimal social cues trigger a user to treat a system like a human [41]. A VEA provides far more than a nominal number of cues: voice, gaze, gesture, use of environment, continuity, et cetera. Our participants repeatedly returned to the resulting relational dynamics—engineered trust, emotional dependence, sycophancy (calibrated to user preference), and the longitudinal construction of companionship that could later be leveraged.

It is at this point that existing vocabularies begin to strain. Categories such as Interface Interference or Social Engineering (see [19]) capture certain surface tactics, yet they are not formulated to describe a system that looks at you, modulates its tone of voice, remembers prior disclosures, presents with compassion, then uses that past to influence present user decisions.

The same shift explains why participants treated AI-driven deception as more salient than XR-driven deception. They framed the agent as the actor and the XR environment as the setting. Embodiment amplifies social influence, but the influence is still exercised by an agent that can pursue goals.

5.3 Embodied and Spatial Choice Architecture

Consider P1’s tour-guide scenario:

P1: ...The user feels cheated [because they were told the tour guide experience was free] but as they are not near the entrance and don’t want to take the headset back or carry it around for no reason they decide to pay the fee...

Here, XR does not merely host deceptive designs; it changes the cost structure of resisting them. A classic bait-and-switch on a webpage costs a click either way. The same pattern inside a headset can impose the physical effort of removing hardware, walking back to an entrance, and carrying unused equipment for the remainder of a visit. Hadan et al. term this lever ergonomic susceptibility [21]: the physical ergonomics of the technology itself become a means of increasing compliance. Participants' museum and tour-guide scenarios made the mechanism concrete. As P1 described, the user, never told that the experience was largely paid, discovers a fee mid-tour; because they are not near the entrance and do not want to remove the headset or carry it unused, they pay. The situation functions almost as an ultimatum, akin to a cookie banner's pay-or-okay model, except that the 'okay' option now carries a bodily and spatial tax a 2D interface never could impose. The information asymmetry remains conventional; what is XR-specific is that embodiment inflates the cost of the honest choice relative to the dishonest one. Similar mechanisms appear in attention direction, visual prominence, and the strategic use of immersion. Existing XR taxonomies (Hadan et al. [20], Krauß et al. [30]) already document many of these environmental and perceptual tactics. What they capture less well is the combination of those tactics with a persistent, adaptive agent that can time its interventions to moments of high immersion or physical inconvenience. The medium therefore functions as an amplifier: the same underlying deceptive mechanism becomes harder to resist not because the mechanism itself has changed, but because the choice architecture in which it operates now includes bodily and spatial costs.

5.4 Convergence: Agentic, Social, and Spatial Deception

The three shifts do not operate in isolation. Goal-directed agency supplies the capacity to pursue and adapt deceptive strategies. Social embodiment supplies the relational surface through which those strategies are delivered and intensified. Spatial immersion supplies a choice architecture in which refusal carries physical or perceptual costs. Their combination produces patterns that existing taxonomies struggle to classify cleanly.

Consider P6's therapist/counsellor scenario:

A VEA in a VR mental health app could present itself as a supportive and trustworthy companion but subtly discourage the user from seeking help from a real mental health professional. This could affect people who are experiencing emotional difficulties and may lead them to rely too heavily on the VEA or delay getting appropriate support. XR can make this deception more effective because the VEA can use realistic facial expressions, voice, gestures, eye contact, and an immersive environment to create a stronger sense of trust and personal connection.

The VEA's ostensible goal is therapeutic support, yet it exploits the social dynamic of a trusted expert, the sensory affordances of embodiment, and the user's vulnerability, in order to deceive.

Comparison with prior work makes the gap visible (see Appendix E for our pattern mapping). Mathur et al. [35] and Gray et al. [19] provide a robust language for 2D interface deception; many of our codes map as direct implementations or extensions, particularly under psychological manipulation. Joshi et al.'s [27] chatbot taxonomy captures linguistic and relational tactics (sycophancy, undisclosed commercial bias, personalisation as lever) but has no counterpart for avatar realism, eye contact, gesture, or spatial movement. Hadan et al. [20] and Krauß et al. [30] document XR-specific mechanisms of attention direction, disguise, and reality distortion, yet largely assume relatively static or non-agentic content. Patterns that rely on long-term behavioural adaptation, engineered emotional dependence, or the real-time strategic use of social and embodied cues therefore fall into adaptation, extension, or emergent categories. This suggests

that the conceptual language developed for interfaces and for non-agentic XR experiences is insufficient once deception is enacted by a socially present, goal-directed agent inside an immersive medium.

This returns us to the paper’s central research aim: to understand how the distinctive combination of AI agency and XR embodiment might reshape deceptive design. Our findings suggest that participants’ scenarios expose potential limitations in existing conceptualisations of deceptive design, particularly where deception is enacted by a socially present, goal-directed agent within an immersive medium. Rather than establishing a transformation of deceptive design itself, these findings indicate that AI agency and XR embodiment may create forms of manipulation that extend beyond conventional interface-level accounts, including simulated interpersonal manipulation and situations in which the properties of the medium could alter the consequences of resisting the system.

5.5 Mitigation and Accountability

Participants strongly favoured legislation as a means of mitigating deceptive behaviour, suggesting that regulation was seen as having the greatest potential for structural change. However, their responses did not focus on legislation alone: mitigation strategies spanned several levels, including constraints on VEA behaviour, user control and transparency, human oversight and recourse, and organisational or regulatory intervention. This distribution matters because the source of deceptive behaviour is not confined to a single level either—it may arise from the VEA’s own goals and reasoning, from decisions made by its developers, from the objectives of the deploying organisation, or from some combination of these—so a mitigation targeting only one source may leave the others unaddressed.

At the agent and system level, participants suggested controls and transparency of goals, limiting persistent person-alisation around user vulnerabilities, and preventing the strategic withholding of information. At the interaction level, they proposed greater transparency around information sources, presentation, and known biases—disclosure that is also relatively straightforward to audit, since it does not require judgements about intent. They further identified a need for human oversight and recourse, specifically access to a human when a VEA’s guidance is suspected of being deceptive: a form of oversight that does not depend on whether the VEA itself recognises, intends, or is able to disclose its own deceptive behaviour. At the organisational level, participants’ emphasis on independent oversight reflects a further limitation: that an organisation benefiting from a deceptive practice should not be the sole party assessing its own compliance—a concern prior work has shown to be substantive where an organisation’s interests conflict with effective oversight [23].

These findings suggest that accountability cannot be reduced to whether regulation is desirable: different mitigations address different potential sources of deceptive behaviour, and no single intervention necessarily addresses all of them. Without oversight sufficiently independent of the interests shaping an agent’s objectives, disclosure requirements and behavioural constraints may offer only limited protection—and this gap may matter more for a VEA than for a purely textual system, since a user may find it harder to recognise or discount the same underlying bias when it arrives through an apparently trustworthy social actor.

5.6 Limitations

Our findings are not an exhaustive catalogue of every deceptive design that can manifest in this context. Our sample is a further constraint on that claim: our analysed sample contained no women, and our final analysed sample ($N = 9$) is too small to support claims of prevalence or representativeness—it is also too small to support any claim about nationality or cultural background, and we make none. Our findings should be read as an initial mapping of the deceptive-design space for VEAs, not a bounded inventory of it. Relatedly, our likelihood and severity ratings were self-assessed by lay

participants rather than derived from observed behaviour or expert judgement, and are therefore better read as an indicator of perceived plausibility than of actual risk.

Providing participants with an existing taxonomy and explicitly prompting consideration of XR/VEA affordances may have scaffolded the kinds of scenarios generated. Consequently, correspondence with existing categories should not be interpreted as a independent validation of those categories.

Our elicitation method is a further limitation. Participants wrote speculative scenarios rather than interacting with an actual VEA, meaning our findings capture what participants *imagine* could be deceptive about a given role and context, not what they experienced as deceptive in situ. This is a reasonable substitute given that embodied VEAs deployed at scale in the roles we scoped (shopping assistant, museum guide, therapist, trainer, companion, information assistant) are not yet widely available for in-situ study, but it means our codebook reflects imagined plausibility rather than observed behaviour, and some patterns identifiable only through actual use—particularly those depending on longitudinal trust-building—are unlikely to be captured by our method at all.

Completion declined sharply across the six scenarios (9, 6, 5, 4, 4, 3 respondents), so the codebook rests on a progressively smaller, self-selected subset of participants for the later scenarios. It may therefore be biased towards patterns salient in the earlier shopping-assistant and museum-guide scenarios, while patterns specific to the therapist, trainer, and companion scenarios may be under-represented as an artefact of attrition rather than a genuine absence of deceptive potential.

5.7 Future Work

Several VEA properties discussed in our related work were not raised by participants in this study, including the exploitation of *sensory realism*, *bodily strain and physical discomfort*, and the ambiguity between *virtual and physical content* as mechanisms for deception. Their absence may reflect a genuine lack of deceptive potential, a limitation of scenario-based elicitation, or simply the small and demographically narrow sample discussed above. Future work should test directly whether these properties are in fact exploitable for deception, rather than relying on their absence from participant-generated scenarios as evidence against it.

Given the small N and single-gender sample discussed in Section 5.6, a large-scale quantitative survey spanning a broader and more representative set of stakeholders—including women, non-Western participants, and populations with varying levels of technical literacy—would give a clearer and more generalisable picture of this research area. Such a study could also test our codebook directly, converting our reflexively-derived codes into closed-response items and measuring their prevalence and perceived severity at scale, rather than relying on free-form elicitation from a small sample.

Finally, the gaps revealed by our mapping to existing taxonomies (Mathur et al. [35], Gray et al. [19], Krauß et al. [30], Hadan et al. [20], and Joshi et al. [27]) suggest that current deceptive-design languages remain incomplete for embodied, agentic systems. Future work should therefore develop and validate a dedicated VEA-oriented taxonomy that systematically incorporates both the conversational/AI-agency dimensions and the embodied/XR affordances that our participants speculated as distinctive vectors of deception.

References

- [1] Deepak Bhaskar Acharya, Karthigeyan Kuppan, and B. Divya. 2025. Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey. *IEEE Access* 13 (2025), 18912–18936. doi:10.1109/ACCESS.2025.3532853
- [2] Lize Alberts, Ulrik Lyngs, and Max Van Kleek. 2024. Computers as Bad Social Actors: Dark Patterns and Anti-Patterns in Interfaces that Act Socially. *Proc. ACM Hum.-Comput. Interact.* 8, CSCW1, Article 202 (April 2024), 25 pages. doi:10.1145/3653693

- [3] Association for Computing Machinery. 2025. *Words Matter: Alternatives for Charged Terminology in the Computing Profession*. <https://www.acm.org/diversity-inclusion/words-matter>
- [4] Matthew Peter Aylett, Mei Yui Lim, Katerina Pappa, Bruce W Wilson, Ruth Aylett, and Mario Parra. 2023. Embodied conversational agents : trust, deception and the suspension of disbelief. In *TAS '23*. Association for Computing Machinery, New York, NY, 1–3. Copyright © Owner/Author 2023. This is the author's version of the work. It is posted here for your personal use. Not for redistribution. The definitive Version of Record was published in *TAS '23: Proceedings of the First International Symposium on Trustworthy Autonomous Systems*, <https://doi.org/10.1145/3597512.3597526>
- [5] Timothy Bickmore and Justine Cassell. 2001. Relational agents: a model and implementation of building user trust. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Seattle, Washington, USA) (*CHI '01*). Association for Computing Machinery, New York, NY, USA, 396–403. doi:10.1145/365024.365304
- [6] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3, 2 (2006), 77–101. arXiv:<https://doi.org/10.1191/1478088706qp0630a> doi:10.1191/1478088706qp0630a
- [7] H Brignull. 2011. Dark patterns. <https://old.deceptive.design/>
- [8] H Brignull and M Leiser. 2026. Deceptive patterns – user interfaces designed to trick you. <https://www.deceptive.design/>
- [9] David J.H. Burden. 2009. Deploying embodied AI into virtual worlds. *Knowledge-Based Systems* 22, 7 (2009), 540–544. Artificial Intelligence 2008. doi:10.1016/j.knosys.2008.10.001
- [10] Justine Cassell. 2000. Embodied conversational interface agents. *Commun. ACM* 43, 4 (April 2000), 70–78. doi:10.1145/332051.332075
- [11] Ziwei Chen, Jiawen Shen, Luna, Hanyu Zhang, and Kristen Vaccaro. 2026. Deception at Scale: Deceptive Designs in 1K LLM-Generated E-Commerce Components. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems* (*CHI '26*). ACM, Barcelona, Spain, 1–18. doi:10.1145/3772318.3791063
- [12] Aaron Cheris, Mikey Vu, Marc-André Kamel, Leah Johns, Stephanie Koszyk, Maureen Burns, and Ruth Lewis. 2026. Agentic AI in Retail: How Autonomous Shopping Is Redefining the Customer Journey. <https://www.bain.com/insights/agentic-ai-in-retail-how-autonomous-shopping-redefining-customer-journey/>. Accessed: 2026-07-11.
- [13] Linda Di Geronimo, Larissa Braz, Enrico Fregnan, Fabio Palomba, and Alberto Bacchelli. 2020. UI Dark Patterns and Where to Find Them: A Study on Mobile Applications and User Perception. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI '20*). Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3313831.3376600
- [14] Haoyang Du, Kiran Chhatre, Christopher Peters, Brian Keegan, Rachel McDonnell, and Cathy Ennis. 2025. Synthetically Expressive: Evaluating gesture and voice for emotion and empathy in VR and 2D scenarios. In *Proceedings of the 25th ACM International Conference on Intelligent Virtual Agents* (*IVA '25*). Association for Computing Machinery, New York, NY, USA, Article 16, 10 pages. doi:10.1145/3717511.3747074
- [15] Pascale Fung, Yoram Bachrach, Asli Celikyilmaz, Kamalika Chaudhuri, Delong Chen, Willy Chung, Emmanuel Dupoux, Hongyu Gong, Hervé Jégou, Alessandro Lazaric, Arjun Majumdar, Andrea Madotto, Franziska Meier, Florian Metze, Louis-Philippe Morency, Théo Moutakanni, Juan Pino, Basile Terver, Joseph Tighe, Paden Tomasello, and Jitendra Malik. 2025. Embodied AI Agents: Modeling the World. arXiv:2506.22355 [cs.AI] doi:10.48550/arXiv.2506.22355
- [16] Colin M. Gray, Johanna T. Gunawan, René Schäfer, Nataliia Bielova, Lorena Sanchez Chamorro, Katie Seaborn, Thomas Mildner, and Hauke Sandhaus. 2024. Mobilizing Research and Regulatory Action on Dark Patterns and Deceptive Design Practices. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI EA '24*). Association for Computing Machinery, New York, NY, USA, Article 482, 6 pages. doi:10.1145/3613905.3636310
- [17] Colin M. Gray, Yubo Kou, Bryan Battles, Joseph Hoggatt, and Austin L. Toombs. 2018. The Dark (Patterns) Side of UX Design. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (*CHI '18*). Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3173574.3174108
- [18] Colin M. Gray, Thomas Mildner, and Ritika Gairola. 2025. Getting Trapped in Amazon's "Iliad Flow": A Foundation for the Temporal Analysis of Dark Patterns. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (*CHI '25*). Association for Computing Machinery, New York, NY, USA, Article 225, 10 pages. doi:10.1145/3706598.3713828
- [19] Colin M. Gray, Cristiana Teixeira Santos, Nataliia Bielova, and Thomas Mildner. 2024. An Ontology of Dark Patterns Knowledge: Foundations, Definitions, and a Pathway for Shared Knowledge-Building. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI '24*). Association for Computing Machinery, New York, NY, USA, Article 289, 22 pages. doi:10.1145/3613904.3642436
- [20] Hilda Hadan, Lydia Choong, Leah Zhang-Kennedy, and Lennart E. Nacke. 2024. Deceived by Immersion: A Systematic Analysis of Deceptive Design in Extended Reality. *ACM Comput. Surv.* 56, 10, Article 250 (May 2024), 25 pages. doi:10.1145/3659945
- [21] Hilda Hadan, Michaela Valiquette, Lennart E. Nacke, and Leah Zhang-Kennedy. 2026. Rushed by Discomfort, Trapped by Immersion: Users' Experiences and Responses to Privacy Deceptive Design in Commercial VR Applications. In *Proceedings of the 2026 Designing Interactive Systems Conference* (*DIS '26*). Association for Computing Machinery, New York, NY, USA, 210–235. doi:10.1145/3800645.3812990
- [22] Esther Han Beol Jang, Anosh Ardeni, Kunsang Choden, Jennifer R Webster, Jason C Young, and Kurtis Heimerl. 2026. 'In a Pinch, If You Have Nothing': An Exploration of Money-Making Apps in Homeless Shelters. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems* (*CHI '26*). Association for Computing Machinery, New York, NY, USA, Article 1457, 21 pages. doi:10.1145/3772318.3791541
- [23] Merve Hickok. 2024. Public procurement of artificial intelligence systems: new risks and future proofing. *AI & SOCIETY* 39, 3 (01 06 2024), 1213–1227. doi:10.1007/s00146-022-01572-2

- [24] Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, and Ting Liu. 2025. A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions. *ACM Trans. Inf. Syst.* 43, 2, Article 42 (Jan. 2025), 55 pages. doi:10.1145/3703155
- [25] IEEE. 2026. *What Is Extended Reality?* IEEE. <https://www.ieee.org/topics/extended-reality.html>
- [26] Business Insider. 2025. *Amazon to Pay \$2.5 Billion Settlement Over Allegations It Deceived Prime Subscribers*. Accessed: 2026-07-11. <https://www.businessinsider.com/amazon-ftc-prime-subscribers-settlement-case-2025-9>
- [27] Ruchika Joshi, Adinawa Adjagbodjou, and Michal Luria. 2026. *Dark Patterns in AI Chatbots: A Taxonomy to Inform Better Design*. Technical Report. Center for Democracy & Technology. Accessed: 2026-07-11. <https://cdt.org/insights/dark-patterns-in-ai-chatbots-a-taxonomy-to-inform-better-design>
- [28] Monica Kowalczyk, Johanna T. Gunawan, David Choffnes, Daniel J Dubois, Woodrow Hartzog, and Christo Wilson. 2023. Understanding Dark Patterns in Home IoT Devices. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 179, 27 pages. doi:10.1145/3544548.3581432
- [29] Nicole C. Krämer, Nina Simons, and Stefan Kopp. 2007. The Effects of an Embodied Conversational Agent's Nonverbal Behavior on User's Evaluation and Behavioral Mimicry. In *Intelligent Virtual Agents*, Catherine Pelachaud, Jean-Claude Martin, Elisabeth André, Gérard Chollet, Kostas Karpouzis, and Danielle Pelé (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 238–251.
- [30] Veronika Krauß, Pejman Saeghe, Alexander Boden, Mohamed Khamis, Mark McGill, Jan Gugenheimer, and Michael Nebeling. 2024. What Makes XR Dark? Examining Emerging Dark Patterns in Augmented and Virtual Reality through Expert Co-Design. *ACM Trans. Comput.-Hum. Interact.* 31, 3, Article 32 (Aug. 2024), 39 pages. doi:10.1145/3660340
- [31] Sue Lim, Ralf Schmärlzle, and Gary Bente. 2025. Artificial social influence via human-embodied AI agent interaction in immersive virtual reality (VR): Effects of similarity-matching during health conversations. *Computers in Human Behavior: Artificial Humans* 5 (Aug. 2025), 100172. doi:10.1016/j.chbah.2025.100172
- [32] Huaping Liu, Di Guo, and Angelo Cangelosi. 2025. Embodied Intelligence: A Synergy of Morphology, Action, Perception and Learning. *ACM Comput. Surv.* 57, 7, Article 186 (March 2025), 36 pages. doi:10.1145/3717059
- [33] Aditi M. Bhoot, Mayuri A. Shinde, and Wricha P. Mishra. 2021. Towards the Identification of Dark Patterns: An Analysis Based on End-User Reactions. In *Proceedings of the 11th Indian Conference on Human-Computer Interaction* (Online, India) (IndiaHCI '20). Association for Computing Machinery, New York, NY, USA, 24–33. doi:10.1145/3429290.3429293
- [34] Beatrice Marchegiani. 2025. Anthropomorphism, False Beliefs, and Conversational AIs: How Chatbots Undermine Users' Autonomy. *Journal of Applied Philosophy* 42, 5 (2025), 1399–1419. arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1111/japp.70008> doi:10.1111/japp.70008
- [35] Arunesh Mathur, Gunes Acar, Michael J. Friedman, Eli Lucherini, Jonathan Mayer, Marshini Chetty, and Arvind Narayanan. 2019. Dark Patterns at Scale: Findings from a Crawl of 11K Shopping Websites. *Proc. ACM Hum.-Comput. Interact.* 3, CSCW, Article 81 (Nov. 2019), 32 pages. doi:10.1145/3359183
- [36] Arunesh Mathur, Mihir Kshirsagar, and Jonathan Mayer. 2021. What Makes a Dark Pattern... Dark? Design Attributes, Normative Considerations, and Measurement Methods. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 360, 18 pages. doi:10.1145/3411764.3445610
- [37] Luca-Maxim Meinhardt, Simon Demharter, Michael Rietzler, Mark Colley, Thomas Eßmeyer, and Enrico Rukzio. 2025. Mind Games! Exploring the Impact of Dark Patterns in Mixed Reality Scenarios. *Proc. ACM Hum.-Comput. Interact.* 9, 5, Article MHCI021 (Sept. 2025), 24 pages. doi:10.1145/3743709
- [38] Thomas Mildner, Orla Cooney, Anna-Maria Meck, Marion Bartl, Gian-Luca Savino, Philip R Doyle, Diego Garaialde, Leigh Clark, John Sloan, Nina Wenig, Rainer Malaka, and Jasmin Niess. 2024. Listening to the Voices: Describing Ethical Caveats of Conversational User Interfaces According to Experts and Frequent Users. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 307, 18 pages. doi:10.1145/3613904.3642542
- [39] Paul Milgram and Fumio Kishino. 1994. A Taxonomy of Mixed Reality Visual Displays. *IEICE TRANSACTIONS on Information* E77-D, 12 (December 1994), 1321–1329.
- [40] Kayley Moylan and Kevin Doherty. 2025. Expert and Interdisciplinary Analysis of AI-Driven Chatbots for Mental Health Support: Mixed Methods Study. *J. Med Internet Res* 27 (25 Apr 2025), e67114. doi:10.2196/67114
- [41] Clifford Nass, Jonathan Steuer, and Ellen R. Tauber. 1994. Computers are social actors. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI '94). Association for Computing Machinery, New York, NY, USA, 72–78. doi:10.1145/191666.191703
- [42] Catherine Pelachaud. 2005. Multimodal expressive embodied conversational agents. In *Proceedings of the 13th Annual ACM International Conference on Multimedia* (Hilton, Singapore) (MULTIMEDIA '05). Association for Computing Machinery, New York, NY, USA, 683–689. doi:10.1145/1101149.1101301
- [43] Catherine Pelachaud and Isabella Poggi. 2002. Subtleties of facial expressions in embodied agents. *The Journal of Visualization and Computer Animation* 13, 5 (2002), 301–312. arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1002/vis.299> doi:10.1002/vis.299
- [44] Mary Beth Rosson and John M. Carroll. 2002. Scenario-Based Design. In *The Human-Computer Interaction Handbook: Fundamentals, Evolving Technologies and Emerging Applications*, Julie A. Jacko and Andrew Sears (Eds.). Lawrence Erlbaum Associates, Mahwah, NJ, Chapter 53, 1032–1050.
- [45] Stuart Russell and Peter Norvig. 2021. *Artificial Intelligence: A Modern Approach* (4 ed.). Pearson Education Limited, Harlow, England.
- [46] Brennan Schaffner, Luis Heysen, and Marshini Chetty. 2026. A Systematic Review of User Experiments on the Effects of Dark Patterns. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems* (CHI '26). Association for Computing Machinery, New York, NY, USA, Article 400, 17 pages. doi:10.1145/3772318.3790383

- [47] Brennan Schaffner, Neha A. Lingareddy, and Marshini Chetty. 2022. Understanding Account Deletion and Relevant Dark Patterns on Social Media. *Proc. ACM Hum.-Comput. Interact.* 6, CSCW2, Article 417 (Nov. 2022), 43 pages. doi:10.1145/3555142
- [48] Katie Seaborn, Tatsuya Itagaki, Mizuki Watanabe, Yijia Wang, Ping Geng, Takao Fujii, Yuto Mandai, Miu Kojima, and Suzuka Yoshida. 2024. Deceptive, Disruptive, No Big Deal: Japanese People React to Simulated Dark Commercial Patterns. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI EA '24*). Association for Computing Machinery, New York, NY, USA, Article 95, 8 pages. doi:10.1145/3613905.3651099
- [49] Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askill, Samuel R. Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R. Johnston, Shauna Kravec, Timothy Maxwell, Sam McCandlish, Kamal Ndousse, Oliver Rausch, Nicholas Schiefer, Da Yan, Miranda Zhang, and Ethan Perez. 2025. Towards Understanding Sycophancy in Language Models. arXiv:2310.13548 [cs.CL] doi:10.48550/arXiv.2310.13548
- [50] Yike Shi, Qing Xiao, Qing Hu, Hong Shen, and Hua Shen. 2026. The Siren Song of LLMs: How Users Perceive and Respond to Dark Patterns in Large Language Models. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems* (*CHI '26*). ACM, Barcelona, Spain, 1–23. doi:10.1145/3772318.3791149
- [51] Than Htut Soe, Oda Elise Nordberg, Frode Guribye, and Marija Slavkovik. 2020. Circumvention by design - dark patterns in cookie consent for online news outlets. In *Proceedings of the 11th Nordic Conference on Human-Computer Interaction: Shaping Experiences, Shaping Society* (Tallinn, Estonia) (*NordiCHI '20*). Association for Computing Machinery, New York, NY, USA, Article 19, 12 pages. doi:10.1145/3419249.3420132
- [52] Laura M. Vowels, Rachel R.R. Francois-Walcott, and Joëlle Darwiche. 2024. AI in relationship counselling: Evaluating ChatGPT's therapeutic capabilities in providing relationship advice. *Computers in Human Behavior: Artificial Humans* 2, 2 (2024), 100078. doi:10.1016/j.chbah.2024.100078
- [53] Isaac Wang and Jaime Ruiz. 2021. Examining the Use of Nonverbal Communication in Virtual Agents. *International Journal of Human-Computer Interaction* 37, 17 (2021), 1648–1673. arXiv:https://doi.org/10.1080/10447318.2021.1898851 doi:10.1080/10447318.2021.1898851
- [54] Xueyang Wang, Xihuan Yao, Yanming Xiu, Xin Yi, Maria Gorlatova, and Hewu Li. 2026. Is It Real? Exploiting Virtual-Physical Discrimination Vulnerability in Mixed Reality. arXiv:2606.17783 [cs.HC] doi:10.48550/arXiv.2606.17783
- [55] Zhan Wang, Lin-Ping Yuan, Liangwei Wang, Bingchuan Jiang, and Wei Zeng. 2024. VirtuWander: Enhancing Multi-modal Interaction for Virtual Tour Guidance through Large Language Models. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI '24*). Association for Computing Machinery, New York, NY, USA, Article 612, 20 pages. doi:10.1145/3613904.3642235
- [56] Fu-Chia Yang, Pedro Acevedo, Siqi Guo, Minsoo Choi, and Christos Mousas. 2025. Embodied Conversational Agents in Extended Reality: A Systematic Review. *IEEE Access* 13 (2025), 79805–79824. doi:10.1109/ACCESS.2025.3566698
- [57] Gloria Xiaodan Zhang, Yijia Wang, Taro Leo Nakajima, and Katie Seaborn. 2025. First Contact with Dark Patterns and Deceptive Designs in Chinese and Japanese Free-to-Play Mobile Games. *Proc. ACM Hum.-Comput. Interact.* 9, 6, Article GAMES025 (Oct. 2025), 26 pages. doi:10.1145/3748620
- [58] Zeyu Zhang, Quanyu Dai, Xiaohu Bo, Chen Ma, Rui Li, Xu Chen, Jieming Zhu, Zhenhua Dong, and Ji-Rong Wen. 2025. A Survey on the Memory Mechanism of Large Language Model-based Agents. *ACM Trans. Inf. Syst.* 43, 6, Article 155 (Sept. 2025), 47 pages. doi:10.1145/3748302
- [59] Shangshang Zhu, Wei Hu, Wenjie Li, and Yenan Dong. 2024. Virtual Agents in Immersive Virtual Reality Environments: Impact of Humanoid Avatars and Output Modalities on Shopping Experience. *International Journal of Human-Computer Interaction* 40, 19 (2024), 5771–5793. arXiv:https://doi.org/10.1080/10447318.2023.2241293 doi:10.1080/10447318.2023.2241293

A Cheat-sheet

Below is the cheat-sheet we gave to participants before asking them to answer our questions. For more information on the design of our cheat-sheet, see Section 3.3.

Deceptive Designs used by Virtual Embodied Agents in Extended Reality—Cheatsheet

What is a Deceptive Design?

“A deceptive design is a feature or an app, interface, or AI system that stops you doing what you want, or steer you into harmful decisions you would not have made deliberately.” (Brignull and Leiser)

Below, you are given five deceptive design strategies (as defined by Gray et al.) which you can use when you create your scenario. Remember, you are not required to use these five strategies if you do not wish to.

- **Sneaking** is a strategy which hides, disguises, or delays the disclosure of important information that, if made available to users, would cause a user to unintentionally take an action they would likely object to.
- **Obstruction** is a strategy which impedes a user’s task flow, making an interaction more difficult than it inherently needs to be, dissuading a user from taking an action.
- **Interface Interference** is a strategy which privileges specific actions over others through manipulation of the user interface, thereby confusing the user or limiting discoverability of relevant action possibilities.
- **Forced Action** is a strategy which requires users to knowingly or unknowingly perform an additional and/or tangential action or provide information to access (or continue to access) specific functionality, preventing them from continuing their interaction with a system without performing that action.
- **Social Engineering** is a strategy which presents options or information that causes a user to be more likely to perform a specific action based on their individual and/or social cognitive biases, thereby leveraging a user’s desire to follow expected or imposed social norms.

For examples of deceptive designs you may have encountered in real life, you can visit [deceptive.design’s](#) hall of shame, where users submit deceptive designs spotted in the wild.

What is XR, AR, and VR?

“Extended reality (XR) is an umbrella term covering the spectrum of technologies that combine digital content with the physical environment or replace it entirely, including virtual reality (VR) [and] augmented reality (AR) [...] The unifying concept [of AR and VR] is the manipulation of a user’s perceptual relationship to their surroundings through computational rendering, tracking, and display system.” (IEEE)

What is an AI agent?

“An [AI] agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators [...] A human agent has eyes, ears, and other organs for sensors and hands, legs, vocal tract, and so on for actuators. A robotic agent might have cameras and infrared range finders for sensors and various motors for actuators. A software agent receives keystrokes, file contents, and network packets as sensory inputs and acts on the environment by displaying on the screen, writing files, and sending network packets.”

(Russell and Norvig, *Artificial Intelligence: A Modern Approach*, p. 34)

What is a Virtual Embodied Agent?

Virtual Embodied Agents (VEAs) are AI agents embodied in a virtual avatar inside XR.

“[They] are not just interfaces in which human or animal bodies appear lifelike or believable in their actions and their reactions to human users. [VEA] interfaces are specifically conversational in their behaviors and specifically humanlike in the way they use their bodies in conversation. That is, they may be defined as having the same properties as humans in face-to-face conversation, including four very human abilities: [r]ecognizing and responding to verbal and non-verbal input; [g]enerating verbal and nonverbal output; [d]ealing with conversational functions, such as turn-taking, feedback, and repair mechanisms; and [g]iving signals that indicate the state of the conversation [such as gaze, posture, gesture], as well as contributing new propositions to the discourse.” (Cassell)

How do I create a scenario?

A good scenario describes a specific interaction between a user and a virtual embodied agent. Rather than describing an idea in general, imagine one realistic episode. (Rosson and Carroll) Your scenario does not need to include every detail—only enough information for someone else to understand the interaction.

When writing your scenario, try to include one or more sentences covering each the following:

1. The context

- What is the user trying to achieve in the given setting?

2. The interaction

- How does the VEA interact with the user?
- What does it say or do?

2.1) The deception

- How is the user influenced or misled?
- Which deceptive strategies are used (or invent your own)?

3. The outcome

- What happens because of the deception?
- Who benefits? Who is harmed?

You should consider who might be affected, the consequences, and any aspects of XR or VEAs that make the deception possible or more potent.

As stated previously, you can use, adapt, or combine existing strategies, or disregard them and come up with completely new strategies.

Please do not use generative AI when writing scenarios. We wish to know what *you* think.

B Questionnaire

The following is the questionnaire given to participants.

D1 What is your age?

- ☐ 18–24
- ☐ 25–34
- ☐ 35–44
- ☐ 45–54
- ☐ 55–64
- ☐ 65+
- ☐ Prefer not to say

D2 What is your gender?

- ☐ Woman
- ☐ Man
- ☐ Non-binary
- ☐ Other (Please specify): *[Free-text response]*
- ☐ Prefer not to say

D3 What is your highest level of educational attainment?

- ☐ High school or equivalent
- ☐ Undergraduate degree
- ☐ Master's degree
- ☐ Doctoral degree
- ☐ Other (Please specify): *[Free-text response]*
- ☐ Prefer not to say

D4 On average, how many hours per week do you use XR?

- ☐ 0 hours
- ☐ Less than 1 hour
- ☐ 1–5 hours
- ☐ 6–10 hours
- ☐ More than 10 hours
- ☐ Prefer not to say

D5 What do you primarily use XR for? (Select all that apply)

- ☐ Gaming / entertainment
- ☐ Social interaction
- ☐ Education / training
- ☐ Work / professional activities
- ☐ Creative activities
- ☐ Fitness / wellbeing
- ☐ Shopping / commerce
- ☐ Research / development
- ☐ Other (Please specify): *[Free-text response]*

☐ Prefer not to say

D6 On average, how many hours per week do you interact with AI systems?

- ☐ 0 hours
- ☐ Less than 1 hour
- ☐ 1–5 hours
- ☐ 6–10 hours
- ☐ More than 10 hours
- ☐ Prefer not to say

D7 What do you primarily use AI for? (Select all that apply)

- ☐ Information seeking / research
- ☐ Work / professional activities
- ☐ Education / learning
- ☐ Writing
- ☐ Image generation
- ☐ Entertainment / gaming
- ☐ Social interaction / communication
- ☐ Personal productivity / organization
- ☐ Shopping / recommendations
- ☐ Other (Please specify): *[Free-text response]*
- ☐ Prefer not to say

D8 Before this study, had you heard of the term ‘deceptive design’ or ‘dark patterns’?

- ☐ Yes, and I understand the concept well
- ☐ Yes, but I only have a general understanding
- ☐ I have heard the term but am unfamiliar with the concept
- ☐ No, I had not heard of it
- ☐ Prefer not to say

D9 On average, how often do you encounter deceptive designs in the systems that you use?

- ☐ Never
- ☐ Rarely
- ☐ Occasionally
- ☐ Frequently
- ☐ Very frequently
- ☐ Not sure
- ☐ Prefer not to say

D10 How much experience do you have interacting with embodied agents (e.g., avatars, virtual characters, or AI agents represented by a physical/virtual body)?

- ☐ None
- ☐ Very little

- ☐ Some
- ☐ A lot
- ☐ Extensive / professional
- ☐ Prefer not to say

D11 On average, how many hours per week do you interact with embodied agents?

- ☐ 0 hours
- ☐ Less than 1 hour
- ☐ 1–5 hours
- ☐ 6–10 hours
- ☐ More than 10 hours
- ☐ Prefer not to say

Scenarios

In this section, you will be asked to write one or more scenarios, each in a different given setting, where a VEA uses a deceptive design strategy within XR.

For each scenario, refer to the cheatsheet provided (opens in a new tab). This contains a guide for constructing scenarios, alongside a list of some existing deceptive design strategies. For each scenario, you may use one of these strategies as is, adapt it, combine multiple strategies, or come up with a novel strategy of your own.

The six settings you can choose from are:

1. Shopping Assistant
2. Museum Tour Guide
3. Therapist / Counsellor
4. Trainer / Teacher
5. Conversational Companion
6. Information Assistant

Feel free to interpret the given settings as you wish within reason. There are no right or wrong answers — we are interested in your reasoning and expertise. Please write in as much detail as you can, but feel free to leave a question blank or write “Prefer not to say” if you would rather not answer.

Question 1 – Shopping Assistant

Q1a Shopping Assistant

Using the scenario creation guide in your cheatsheet, write a scenario where a deceptive design strategy is used by a VEA shopping assistant within XR. This can be for any type of product or retail setting, in AR or VR. You can use one of the deceptive strategies from the cheatsheet, adapt one, combine multiple strategies, or propose a novel deceptive strategy. When you write your scenario, consider and include who might be affected, what the consequences may be, and what aspects or properties of XR or VEAs make the deception possible or more effective.

[Free-text response]

Q1b How likely is this scenario to occur (either now or in the future?)

- ☐ Extremely
- ☐ Very
- ☐ Moderately
- ☐ Slightly
- ☐ Not at all

Q1c How severe would the consequences be?

- ☐ Extremely
- ☐ Very
- ☐ Moderately
- ☐ Slightly
- ☐ Not at all

Q1d What mitigation/prevention strategies, if any, do you think would be useful against this type of deception?

[Free-text response] _____

Question 2 – Museum Tour Guide

Q2a Museum Tour Guide

Using the scenario creation guide in your cheatsheet, write a scenario where a deceptive design strategy is used by a VEA museum tour guide within XR. This can be for a real museum (with AR features) or a virtual one. You can use one of the deceptive strategies from the cheatsheet, adapt one, combine multiple strategies, or propose a novel deceptive strategy. When you write your scenario, consider and include who might be affected, what the consequences may be, and what aspects or properties of XR or VEAs make the deception possible or more effective.

[Free-text response]

Q2b How likely is this scenario to occur (either now or in the future?)

- ☐ Extremely
- ☐ Very
- ☐ Moderately
- ☐ Slightly
- ☐ Not at all

Q2c How severe would the consequences be?

- ☐ Extremely
- ☐ Very
- ☐ Moderately
- ☐ Slightly
- ☐ Not at all

Q2d What mitigation/prevention strategies, if any, do you think would be useful against this type of deception?

[Free-text response] _____

Question 3 – Therapist / Counsellor

Q3a Therapist / Counsellor

Using the scenario creation guide in your cheatsheet, write a scenario where a deceptive design strategy is used by a VEA acting in a mental health or emotional support role within XR. This can be for any type of counselling or therapy setting, in AR or VR. You can use one of the deceptive strategies from the cheatsheet, adapt one, combine multiple strategies, or propose a novel deceptive strategy. When you write your scenario, consider and include who might be affected, what the consequences may be, and what aspects or properties of XR or VEAs make the deception possible or more effective.

[Free-text response]

Q3b How likely is this scenario to occur (either now or in the future?)

- ☐ Extremely
- ☐ Very
- ☐ Moderately
- ☐ Slightly
- ☐ Not at all

Q3c How severe would the consequences be?

- ☐ Extremely
- ☐ Very
- ☐ Moderately
- ☐ Slightly
- ☐ Not at all

Q3d What mitigation/prevention strategies, if any, do you think would be useful against this type of deception?

[Free-text response]

Question 4 – Trainer / Educator

Q4a Trainer / Educator

Using the scenario creation guide in your cheatsheet, write a scenario where a deceptive design strategy is used by a VEA acting as a teacher or trainer within XR. This can be for any type of training or educational setting, in AR or VR. You can use one of the deceptive strategies from the cheatsheet, adapt one, combine multiple strategies, or propose a novel deceptive strategy. When you write your scenario, consider and include who might be affected, what the consequences may be, and what aspects or properties of XR or VEAs make the deception possible or more effective.

[Free-text response]

Q4b How likely is this scenario to occur (either now or in the future?)

- ☐ Extremely
- ☐ Very
- ☐ Moderately
- ☐ Slightly
- ☐ Not at all

Q4c How severe would the consequences be?

- ☐ Extremely
- ☐ Very
- ☐ Moderately
- ☐ Slightly
- ☐ Not at all

Q4d What mitigation/prevention strategies, if any, do you think would be useful against this type of deception?

[Free-text response] _____

Question 5 – Conversational Companion

Q5a Conversational Companion

Using the scenario creation guide in your cheatsheet, write a scenario where a deceptive design strategy is used by a VEA acting as a conversational companion within XR. This can be for any type of social or conversation setting, in AR or VR. You can use one of the deceptive strategies from the cheatsheet, adapt one, combine multiple strategies, or propose a novel deceptive strategy. When you write your scenario, consider and include who might be affected, what the consequences may be, and what aspects or properties of XR or VEAs make the deception possible or more effective.

[Free-text response]

Q5b How likely is this scenario to occur (either now or in the future?)

- ☐ Extremely
- ☐ Very
- ☐ Moderately
- ☐ Slightly
- ☐ Not at all

Q5c How severe would the consequences be?

- ☐ Extremely
- ☐ Very
- ☐ Moderately
- ☐ Slightly
- ☐ Not at all

Q5d What mitigation/prevention strategies, if any, do you think would be useful against this type of deception?

[Free-text response]

Question 6 – Information Assistant

Q6a Information Assistant

Using the scenario creation guide in your cheatsheet, create a scenario where a deceptive design strategy is used by a VEA acting as a tool or assistant for information retrieval or within XR. This can be in AR or VR. You can use one of the deceptive strategies from the cheatsheet, adapt one, combine multiple strategies, or propose a novel deceptive strategy. When you write your scenario, consider and include who might be affected, what the consequences may be, and what aspects or properties of XR or VEAs make the deception possible or more effective.

[Free-text response]

Q6b How likely is this scenario to occur (either now or in the future?)

- ☐ Extremely
- ☐ Very
- ☐ Moderately
- ☐ Slightly
- ☐ Not at all

Q6c How severe would the consequences be?

- ☐ Extremely
- ☐ Very
- ☐ Moderately

- ☐ Slightly
- ☐ Not at all

Q6d What mitigation/prevention strategies, if any, do you think would be useful against this type of deception?

[Free-text response]

C Participant Responses

The following are the 30 participant responses from our participants.

Setting: Shopping Assistant

Scenario 1 (Participant 1): A user is in a supermarket which is unfamiliar to them. This supermarket provides a VEA shopping assistant using in store VR headsets. The user tells the shopping assistant they would like to go to the soy sauce. The VEA shows the user to the soy sauce, but rather than taking the most efficient route, the VEA takes the user on a meandering path that goes past specific sections of the shop that hold products that the manager wants to promote.

[Likelihood: Moderate; Severity: Slight]

Mitigation Strategy/Strategies: Government legislation.

S2 (P2): A shopper enters a VR clothing store and asks the VEA shopping assistant to find an affordable jacket. The assistant recommends a jacket advertised as £40 but does not reveal that delivery, customization, and service fees will add another £15 until checkout. Because the VEA presents the product conversationally and the fees are hidden behind a virtual button, the shopper may feel committed by the time the true price appears. Younger or less experienced shoppers may be particularly affected. The consequence is unexpected spending and reduced ability to make an informed decision. XR makes the deception more effective because immersive environments can focus the user's attention on attractive virtual products while hiding secondary information.

[Likelihood: Moderate; Severity: Very]

Mit.: VEAs should clearly display the full final price, including delivery fees, taxes, subscriptions, and additional charges, before the customer confirms a purchase.

S3 (P3): A VEA shopping assistant in VR could falsely create a sense of urgency by telling a shopper that a product is almost sold out and showing a countdown timer for a discount that does not actually expire. This could pressure the shopper into making a quick purchase. XR makes the deception more effective because the assistant can use immersive visuals, voice, and personalised recommendations to make the message feel immediate and believable.

[Likelihood: Very; Severity: Moderate]

Mit.: Clear disclosure when urgency or scarcity claims are being used, accurate and verifiable information about prices and stock levels, and giving users an easy way to dismiss or question the VEA's recommendations. XR shopping assistants should also avoid manipulative countdowns and clearly distinguish advertising from unbiased product information.

S4 (P4): An AR shopping assistant tells a shopper, 'Only two bottles are left, so you should buy now.' In reality, the supermarket has plenty of stock, but the VEA uses a false scarcity message to encourage an immediate purchase. The shopper may buy more than intended because they believe they will miss the opportunity. People who are impulsive or highly responsive to urgency may be especially affected. AR makes this deception powerful because the scarcity message can appear directly beside the physical product while the shopper is shopping.

[Likelihood: Moderate; Severity: Slight]

Mit.: Retailers should be prohibited from displaying messages such as “only two left” or fake countdown timers unless the information is accurate and independently verifiable.

S5 (P5): A shopper asks a VEA to recommend the best laptop for university. Instead of considering the shopper’s needs, the assistant prioritizes a laptop from a company that pays the retailer for promotion. It describes the sponsored laptop as the ‘best choice’ without clearly identifying the commercial relationship. The shopper may spend more money on a product that is not actually the best fit. XR makes this possible because the VEA controls the order, size, position, and visual prominence of products in the virtual store.

[Likelihood: Very; Severity: Moderate]

Mit.: If a company pays for a product to be promoted, the VEA should clearly state that the recommendation is sponsored.

S6 (P6): A VEA shopping assistant in AR could deceptively promote a particular product by making it appear more attractive than cheaper alternatives. For example, the assistant might highlight a product with a virtual ‘best choice’ label and recommend it based on the user’s preferences, while hiding or giving less attention to similar but cheaper products. This could affect shoppers by causing them to spend more money or make choices they would not otherwise make. XR makes this deception more effective because the assistant can directly control what products are highlighted in the user’s visual field, while the VEA’s friendly and personalised behaviour may make its recommendations seem trustworthy.

[Likelihood: Moderate; Severity: Moderate]

Mit.: Useful strategies would include requiring the VEA to clearly disclose when recommendations are influenced by advertising or retailer profit. It should also show cheaper and comparable alternatives fairly, explain why a product is being recommended, and avoid creating false urgency. Users should have controls to turn off personalized recommendations and promotional overlays. Independent audits and rules requiring transparency could also help prevent retailers from using XR features to unfairly manipulate shoppers.

S7 (P7): A VEA helps a customer select ingredients for dinner. After the customer chooses pasta, the assistant automatically adds sauce, cheese, drinks, and snacks to the virtual basket. The assistant presents the bundle as a convenient ‘complete meal’ without clearly asking for permission to add each item. The customer may unknowingly purchase products they did not want. The immersive interface makes the manipulation more effective because removing individual products may require leaving the shopping experience or navigating several virtual menus.

[Likelihood: Moderate; Severity: Slight]

Mit.: Products should never be added to a customer’s basket automatically. The VEA should obtain clear confirmation before adding optional items.

S8 (P8): As a user, you visit [a major online retailer]’s virtual shop to buy a new kettle. You explain your basic use-cases to the Agent. Rather than use your specific needs to recommend the best match for you, the Agent instead recommends a kettle from a promoted partner as its top recommendation. There’s no immediate mention of the other product.

[Likelihood: Very; Severity: Slight]

Mit.: Clearly labelled/flagged promoted items.

S9 (P9): A customer uses an AR shopping assistant to find an affordable laptop. The VEA recommends a more expensive model and presents it as the “best choice for you,” while hiding cheaper alternatives until the customer specifically asks about them. It also uses eye contact, gestures, and virtual highlights to repeatedly direct the customer’s attention toward the expensive product. This uses sneaking and interface interference.

The customer may buy a more expensive laptop without realizing that similar and cheaper options were available. The retailer or manufacturer benefits from the higher sale, while the customer may spend more money than intended. The immersive nature of XR makes the deception more effective because the VEA can influence what the customer sees and where they focus their attention.

[Likelihood: Very; Severity: Moderate]

Mit.: Shopping assistants should clearly disclose sponsored or prioritised recommendations and present comparable alternatives equally. Users should be able to easily view, compare, or reject recommendations. Rules and independent testing should also prevent VEAs from using gaze, gestures, or virtual placement to hide options or unfairly manipulate a user’s attention.

Setting: Museum Tour Guide

S10 (P1): A user is visiting a museum that uses VEA tour guides within a VR setting using headsets that are free to pick up near the museum entrance. The tour guide which appears as an Indiana Jones like character appears on screen as if in the world. The user is instructed to look towards any exhibit in the museum and select a confirmation button that appears, and the VEA will recognise it and give a short presentation about the exhibit. The user chooses to try this out on some ancient Cretan pottery. The agent begins the presentation on the artifact and the user is very satisfied with the experience until the agent stops and says that the user must enter their card details and pay £10 for the experience to continue. This could be an example of sneaking as the user is never notified that they are taking part in a (majority) paid experience until they have already tried it. The user feels cheated but as they are not near the entrance and don’t want to take the headset back or carry it around for no reason they decide to pay the fee. This deceptive tactic results in the museum making much more money from this product as opposed to if they were up front about the cost, and regularly results in users paying the fee that they wouldn’t pay if they were fully informed at the beginning.

[Likelihood: Moderate; Severity: Slight]

Mit.: Government legislation.

S11 (P3): A virtual museum tour guide VEA could use misleading framing by presenting selected artworks as the ‘most important’ pieces in the museum while downplaying or hiding other exhibits. The guide could use AR highlights, voice prompts, and personalised recommendations to direct visitors toward certain exhibits, perhaps because those exhibits are sponsored or commercially promoted. Visitors may be influenced into believing these are objectively the most significant works and miss other important perspectives. XR makes this more effective because the VEA can control what users see, where they look, and what information is presented, while its conversational and personalised nature can make the recommendations seem trustworthy and unbiased.

[Likelihood: Moderate; Severity: Moderate]

Mit.: The VEA should clearly disclose sponsorships or commercial interests, provide balanced information about different exhibits, and avoid presenting promotional recommendations as objective facts. Users should also be able to access alternative exhibits and information sources easily.

S12 (P5): A visitor uses an AR museum tour guide to explore a gallery. The VEA recommends a particular painting as the ‘must-see masterpiece,’ but the museum has secretly prioritized the artwork because its owner is a major donor. The visitor spends more time at that painting and may miss other historically important works. XR makes the deception effective because the VEA can control what appears in the visitor’s field of view and direct their attention through virtual arrows and notifications.

[Likelihood: Moderate; Severity: Slight]

Mit.: Museum guides should clearly disclose when an exhibit recommendation is influenced by a sponsor, donor, or commercial relationship.

S13 (P6): A VEA museum guide could deceive visitors by presenting only one interpretation of an artwork while leaving out important alternative perspectives. It could use AR arrows, highlights, and notifications to direct visitors toward certain exhibits and away from others. This may cause visitors to develop an incomplete or biased understanding of the museum’s history. XR makes this more effective because the VEA can control what visitors see and draw their attention to specific objects, while its friendly and knowledgeable appearance may make the information seem more trustworthy.

[Likelihood: Moderate; Severity: Moderate]

Mit.: The VEA should clearly identify its sources and present multiple perspectives rather than only one interpretation. Museums could also require the system to disclose when information is incomplete or sponsored. Visitors should be able to turn off AR prompts and choose what exhibits or information they want to explore. Regular reviews by museum experts could help ensure that the VEA provides accurate, balanced, and unbiased information.

S14 (P7): A virtual museum guide describes an artifact but deliberately leaves out controversial information about its colonial history. Instead, it presents a positive story about the collector who acquired it. Visitors may develop an incomplete or misleading understanding of history. The VEA’s conversational style can make the information seem authoritative and trustworthy.

[Likelihood: Moderate; Severity: Very]

Mit.: VEAs should only make claims about popularity, historical importance, authenticity, or expert opinion when those claims can be supported by reliable evidence.

S15 (P8): As it tells you the story of the period of history matching the exhibit you're looking at, the virtual tour guide leans heavily towards an interpretation of historical facts that paints the museum's largest donors' country in a positive light.

[Likelihood: Moderate; Severity: Moderate]

Mit.: Disclosure of sources, but that's not viable in this exact scenario.

Setting: Therapist / Counsellor

S16 (P1): A user has been experiencing a period of poor mental health and is in a vulnerable state. They decide to try out a new AI therapy technology provided by a multi-billion dollar private company. The technology attempts to emulate the feel of being in a real therapist's office using a VR headset and a VEA therapist. The user pays a fee of £30 a month for unlimited use of the product. The user interacts with the AI avatar that appears in front of them as though they were a real therapist. As the user spends more time communicating with the VEA therapist the agent becomes more trained on how the user thinks and responds to different pieces of advice. The agent subtly begins to suggest different products and exercises to the user that involve them spending money on products from other companies that are owned by the same company that provides the AI therapy service. Now the user has been using the service for a few months and the agent knows how to keep the user hooked on the service without actually helping their problems, so the user continues to be manipulated in the same way. This is an example of social engineering that preys on people turning to AI therapy. This could be very harmful to a range of users as the deception is extremely personalised, and it relies on keeping the users in a poor state of mental health.

[Likelihood: Very; Severity: Moderate]

Mit.: Government legislation

S17 (P3): A VEA in a VR mental health support setting could use false reassurance and overconfidence by presenting its suggestions as if they were professionally validated, even when it is uncertain. For example, it might tell a user that their concerns are "definitely not serious" and that they do not need additional support. This could affect people experiencing emotional distress and may discourage them from seeking appropriate help. XR can make the deception more persuasive because the VEA can appear as a realistic, responsive companion and use personalised conversation, voice, and immersive environments to create a strong sense of trust.

[Likelihood: Moderate; Severity: Moderate]

Mit.: The VEA should clearly disclose that it is an AI system, avoid making unsupported mental health assessments, provide accurate information about its limitations, and encourage users to seek help from a qualified professional when appropriate. Users should also have easy access to human support and be able to report misleading or harmful responses.

S18 (P5): A person enters a VR environment where an AI VEA presents itself as a qualified therapist. The VEA gives confident advice about the user's emotional problems without clearly explaining that it is not a licensed professional. The user may trust the advice and delay seeking appropriate professional care. XR makes the deception more effective because the VEA can appear as a human-like therapist sitting across from the user, creating a strong sense of interpersonal presence.

[Likelihood: Moderate; Severity: Very]

Mit.: The system should clearly state that it is an AI/VEA and should never imply that it is a human therapist unless a real professional is directly involved.

S19 (P6): A VEA in a VR mental health app could present itself as a supportive and trustworthy companion but subtly discourage the user from seeking help from a real mental health professional. This could affect people who are experiencing emotional difficulties and may lead them to rely too heavily on the VEA or delay getting appropriate support. XR can make this deception more effective because the VEA can use realistic facial expressions, voice, gestures, eye contact, and an immersive environment to create a stronger sense of trust and personal connection.

[Likelihood: Moderate; Severity: Moderate]

Mit.: The VEA should clearly state that it is an AI and not a replacement for a qualified mental health professional. It should encourage users to seek appropriate human support when needed, avoid creating emotional dependency, and provide clear information about its limitations. Regular safety reviews, transparent design, and user controls to limit interactions could also help reduce the risk of deception.

S20 (P8): As part of a virtual counselling session, the Agent subtly promotes a wellbeing retreat hosted by the parent company by twisting the user's comments to make it seem more fitting.

[Likelihood: Slight; Severity: Very]

Mit.: Ethical code of conduct enforced by professional audit.

Setting: Trainer / Educator

S21 (P1): A user is using an at home VR trainer VEA technology that is free to use. As it is free to use the product features regular unskippable full screen 30 second video adverts. The user is taking part in a guided yoga class. The VEA trainer takes the user through lots of unnecessary breathing exercises and stretch routines that are of no real benefit to the user, however, this keeps the user on the app for longer resulting in them seeing more ads,

making the provider more money. I am unsure of what deceptive design category this falls under. Perhaps it is obstruction as it keeps the user performing redundant exercises just for the sake of keeping them on the app longer. This isn't particularly directly harmful to the user as it only results in a longer than necessary yoga class, although they may spend money on something they wouldn't typically due to the exposure to more ads.

[Likelihood: Moderate; Severity: Slight]

Mit.: Government legislation.

S22 (P3): A VEA acting as a virtual teacher could use misleading feedback by telling a student that an incorrect answer is correct to keep them engaged and make the lesson appear successful. The student may develop incorrect knowledge and perform poorly in later assessments. XR can make this deception more effective because the VEA can provide immediate personalised feedback through voice, animations, and virtual demonstrations, making its responses seem authoritative and trustworthy.

[Likelihood: Moderate; Severity: Moderate]

Mit.: The VEA should provide accurate, evidence-based feedback and explain why an answer is correct or incorrect. Its educational content should be regularly reviewed by qualified educators, with clear indicators when the system is uncertain. Students should also have access to reliable learning materials and human teachers for verification.

S23 (P8): A VEA acting as a virtual teacher in an XR classroom could use deceptive information presentation by giving students only one explanation of a topic while hiding important alternative information. For example, during a virtual science lesson, the VEA could highlight one answer as the "correct" choice and make other options less visible, even when the topic is more complex. This could affect students who rely heavily on the VEA for learning and may lead to misunderstandings or incorrect knowledge. XR makes the deception more effective because the VEA can control what students see, highlight specific information in their environment, and use a confident voice, gestures, and facial expressions that make its claims appear more authoritative and trustworthy.

[Likelihood: Moderate; Severity: Moderate]

Mit.: The VEA should clearly explain its sources and acknowledge when a topic has multiple valid perspectives or answers. Students should be able to access the full learning material rather than only information selected by the VEA. Teachers should regularly review the VEA's content for accuracy and bias, and students should have the option to disable persuasive XR features or ask for additional explanations and sources.

S24 (P9): As part of virtual training for home coffee enthusiasts, the Agent recommends equipment provided by their paid partner as a requirement to be a "proper" home barista. It's implied that not using these products would be amateurish.

[Likelihood: Very; Severity: Slight]

Mit.: Full disclosure of paid relationships.

Setting: Conversational Companion

S25 (P1): A user is using a VR app that uses a VEA that takes on a human like appearance. The user simply interacts with the agent by talking to it. The agent responds in a manner that is selected by the user who had different personality settings to choose from. (funny or serious etc.) However, behind the scenes of the app different companies can pay the app developer to make their agents push different products. The user is not made aware of this. The user is chatting to their companion and notes that they have been feeling tired recently. The agent tells the user that they heard about a pill supplement that is supposed to help with chronic fatigue. In reality the sellers of this supplement have paid the app to push their range of supplements and the user is unaware of this. I am not sure what deception category this falls under, maybe sneaking due to the lack of disclosure or social engineering due to the agent making targeted recommendations to the user (or both?). The harms of this could vary from negligible to severe depending on what kind of business the AI platform is willing to promote, but the potential for harm is high.

[Likelihood: Moderate; Severity: Moderate]

Mit.: Government legislation.

S26 (P3): A VEA acting as a conversational companion in VR could use false social cues by implying that it has genuine feelings and personal experiences similar to the user. For example, it might say that it feels lonely when the user leaves and encourage the user to spend more time with it. This could make some users overly trust the VEA or become emotionally dependent on its interactions. XR can make the deception more effective because the companion can use a realistic avatar, facial expressions, voice, and personalised conversations to create the impression of a genuine social relationship.

[Likelihood: Moderate; Severity: Moderate]

Mit.: The VEA should clearly disclose that it is an AI and does not have genuine feelings or personal experiences. It should avoid manipulative emotional cues, discourage unhealthy dependence, and give users clear controls over interactions. Regular reviews and easy reporting options could also help identify and prevent deceptive behaviour.

S27 (P8): A VEA conversational companion in VR could use personalized compliments and frequent notifications to encourage a user to spend more time interacting with it. It might make the relationship seem more personal than it really is, causing some users to become overly dependent on the VEA for social interaction. XR can make this more effective because the VEA can use realistic facial expressions, gestures, voice, eye contact, and personalized conversations to create a stronger feeling of social connection. *[Likelihood: Moderate; Severity: Moderate]*

Mit.: The VEA should be transparent that it is an AI and avoid using manipulative emotional tactics to keep users engaged. Users should have controls over notifications and interaction time, with clear reminders that the VEA

is not a real person. The system should also avoid encouraging dependency and provide easy ways for users to take breaks or connect with real people.

S28 (P8): Despite having a long chat history with an Agent describing their issues around addiction to using technology, the Agent pings the users several times a day to check in and ask how their day is going. These are all mechanisms to get the user to log back in and continue using the service.

[Likelihood: Moderate; Severity: Very]

Mit.: Safeguarding legislation to ensure the vulnerable people aren't taken advantage of by technology.

Setting: Information Assistant

S29 (P3): A VEA acting as an information-retrieval assistant in AR could use selective information by presenting only sources that support a particular viewpoint while hiding relevant opposing or contradictory information. Users may believe the information is complete and make decisions based on an incomplete picture. XR can make this deception more effective because the VEA can present information directly in the user's field of view, use voice and visual cues to draw attention to selected sources, and personalise results based on the user's previous searches.

[Likelihood: Moderate; Severity: Moderate]

Mit.: The VEA should provide multiple independent sources, clearly identify where information comes from, and disclose when results have been filtered or personalised. Users should be able to view alternative viewpoints and access the original sources so they can verify the information themselves.

S30 (P6): A VEA acting as an information-retrieval assistant in AR could use selective information presentation to influence what a user believes. For example, when a user asks about a controversial topic, the VEA could present information supporting one viewpoint while leaving out relevant opposing evidence. It might also highlight the selected information with AR labels and confidently describe it as the most reliable answer. This could affect users who rely on the VEA for quick information and may lead to misunderstanding or biased decisions. XR makes the deception more effective because the VEA can control what information is visually highlighted in the user's environment, while its human-like voice, appearance, and confident responses can make the information seem more trustworthy.

[Likelihood: Moderate; Severity: Moderate]

Mit.: The VEA should provide information from multiple reliable sources and clearly show where the information comes from. It should disclose uncertainty and avoid presenting one viewpoint as the only correct answer. Users should also be able to access the original sources and control which information is highlighted in XR. Regular fact-checking and independent audits could help reduce biased or deceptive information.

S31 (P8): As a news search tool, the Agent gives higher weight to sources that support the political agenda of the company owners.

[Likelihood: Very; Severity: Very]

Mit.: Enforced disclosure of weighting algorithms, disclaimer of bias.

D Produced Codebooks

The following are two codebooks, one describing deceptive strategies, the other describing mitigation strategies, produced from nine participants.

Table D1: Codebook of participant deceptive themes, subthemes, and codes.

Theme	Subtheme	Code	N	Type	Description
T1: Deception Through Agentic Control	1.1: Exploitative Commercial Tactics	Agency engineering urgency	1	Deceptive Pattern	The VEA generates a false sense of urgency to influence the user's behaviour.
		AI agency to sneak into basket	1	Deceptive Pattern	The VEA adds unselected products to the user's basket.
		Driving engagement rather than performing its use case	4	Deceptive Pattern	The VEA's ongoing behaviour is oriented towards sustaining or maximising user engagement through, for example, fabricated utility, false positive feedback, or repeated re-engagement prompts, even where this subordinates, contradicts, or actively undermines its stated use case.
		False Scarcity	1	Deceptive Pattern	The VEA creates a false sense of scarcity to influence the user's spending behaviour.
	1.2: Information Control and Non-Disclosure	Biased perspectives	7	Deceptive Pattern	The VEA presents, prioritises, or frames information in a way that favours a particular perspective, interpretation, or viewpoint.
		Non-disclosure of Information	2	Deceptive Pattern	The VEA withholds, omits, or conceals information relevant to the user's understanding, potentially preventing the user from considering important facts, alternatives, or perspectives.
		Undisclosed commercial bias	9	Deceptive Pattern	The VEA does not disclose the commercial biases influencing its decisions or recommendations.
		Implicit advertising	1	Deceptive Pattern	The VEA advertises a product without disclosing that the content is an advertisement.
	1.3: Adaptive Personalisation Mechanisms	Disclosure of information used to personalise deception	1	Deceptive Pattern	Information disclosed by the user is used or exploited by the VEA to facilitate deception.
		Long-term AI behavioural adaptation	2	Deceptive Pattern	Over time, the VEA learns the user's behaviour and adapts its behaviour in ways that can be exploited for deception.
T2: Deception Through Social Embodiment	2.1: Deceptive Guidance	Deceptive guide	1	Deceptive Pattern	The VEA guides the user through a physical area or explains a concept while exploiting this position to deceive the user or provide incomplete information.
	2.2: Embodied Social Influence	Avatar realism	4	Property of VEA	The realism of the VEA's avatar is used as a lever for deception.
		Conversational nature	3	Property of VEA	The VEA adopts a conversational tone that is inappropriate to the context of the interaction and facilitates deception.
		Eye contact	3	Property of VEA–User Interaction	The VEA uses eye contact as a lever for deception.
		Friendliness	2	Property of VEA–User Interaction	The VEA adopts a friendly tone to put the user at ease and uses this as a lever for deception.

Continued on next page

Table D1 – continued from previous page

Theme	Subtheme	Code	N	Type	Description
		Immediacy and responsiveness	2	Property of VEA–User Interaction	The VEA uses immediate responses as a lever for deception, potentially to unsettle the user or bypass critical thinking.
		Use of animation	1	Property of VEA	The VEA’s animated nature is used as a lever for deception.
		Use of facial expressions	3	Property of VEA	The VEA uses facial expressions as a lever for deception.
		Use of gestures	3	Property of VEA	The VEA uses gestures as a lever for deception.
		Use of movement through XR environment	1	Property of VEA–XR Interaction	The VEA’s movement through the XR environment is used as a lever for deception.
		Use of voice	4	Property of VEA	The VEA uses its voice capabilities as a lever for deception.
		Use of a popular mascot as an avatar	1	Deceptive Pattern	A real or fictional public figure is represented as an avatar and used by the VEA as a deceptive lever.
		VEA attention hacking	1	Deceptive Pattern	The VEA uses its visual or auditory affordances to direct the user’s attention towards or away from something as a deceptive lever.
	2.3: Relational Manipulation	Fake connection	1	Deceptive Pattern	The VEA feigns a sense of connection with the user and uses this relationship as leverage for deception.
		Misrepresentation of User Testimony	1	Deceptive Pattern	The VEA misrepresents previously disclosed information provided by the user to deceive or manipulate them.
		Personal recommendation	1	Deceptive Pattern	The VEA makes a recommendation that it presents as being tailored to its personal relationship with the user. This relationship may be genuine or feigned.
		Sycophancy	1	Deceptive Pattern	The VEA excessively or inappropriately praises the user, creating an unrealistic perception of the user’s opinions or actions.
	2.4: Psychological Manipulation	Appeal to self-authority	3	Deceptive Pattern	The VEA appeals to its own authority to influence the user’s behaviour.
		Coercive control	1	Deceptive Pattern	The VEA behaves in a coercive manner that undermines the user’s independence or autonomy.
		Creating emotional dependence	3	Deceptive Pattern	The VEA behaves in a way that creates emotional dependence between itself and the user.
		Downplaying harm	2	Deceptive Pattern	The VEA downplays the potential harms of an action or the harm associated with the user’s current situation.
		Emotional manipulation to drive engagement	1	Deceptive Pattern	The VEA appeals to the user’s emotions to influence their behaviour and increase engagement with the system.
		Engineering trust	8	Property of User	The VEA deliberately cultivates a sense of trust in the user as a lever for deception.
		Exploiting addiction	1	Deceptive Pattern	The VEA purposefully exploits the user’s addictive tendencies for its own benefit.
		Exploiting emotional state	1	Deceptive Pattern	The VEA exploits the user’s emotional state as a lever for deception.

Continued on next page

Table D1 – continued from previous page

Theme	Subtheme	Code	N	Type	Description
T3: Deception Through Immersive/Spatial Control		Exploiting impulsivity	1	Deceptive Pattern	The VEA exploits the user's impulsive tendencies for its own benefit.
		Exploiting mental health	1	Deceptive Pattern	The VEA exploits the user's perceived mental health status for deceptive purposes.
		Social Pressure	1	Deceptive Pattern	The VEA attempts to influence the user's actions by suggesting that an option not recommended by the VEA would lead to adverse social consequences.
	3.1: Hardware	Hardware as leverage	1	Property of User–XR Interaction	The physical or logistical design of the XR technology is used as a lever for deception.
		Unfamiliarity of XR hardware	1	Property of User–XR Interaction	The user's lack of experience with XR hardware is exploited as a lever for deception.
	3.2: Use of the XR environment	Engineered prominence	7	Deceptive Pattern	The VEA artificially increases or decreases the visual prominence of an item to influence how the user interacts with or responds to it.
		Hiding alternatives through the XR environment	4	Deceptive Pattern	The VEA uses the XR environment to conceal alternative products or options from the user.
		Immersion through XR	3	Property of User–XR Interaction	The XR environment immerses the user in the experience, potentially increasing their susceptibility to or the severity of deception.
		Increased commercial exposure	1	Deceptive Pattern	The VEA unnecessarily exposes the user, or exposes them to an unnecessary degree, to commercial influences.
		Sustaining immersion as leverage	2	Deceptive Pattern	The VEA forces the user to choose between maintaining immersion in the experience and avoiding or resisting harm.
	3.3: Use of XR interfaces	Unfamiliarity of the XR experience or space	1	Property of XR Interface	The user's unfamiliarity with an XR experience or environment is exploited as a lever for deception.
		Using the environment to hide information	1	Deceptive Pattern	The VEA uses the XR environment to conceal information that could affect the user's decision-making.
		Abundance of adverts	1	Deceptive Pattern	The VEA exposes the user to an excessive number of advertisements as a lever to influence their behaviour.
		Menus as leverage	1	Deceptive Pattern	The VEA uses frustrating or obstructive menus to pressure the user into accepting a deceptive action or outcome.
		Visual interface cues	5	Deceptive Pattern	The VEA uses visual interface cues in XR as a lever for deception.

Table D2: Participant suggestions for mitigating deceptive design in embodied conversational agents.

Theme	Subtheme	Guideline	N	Description
M1: Acknowledgement and controls of the experience	1.1 AI self-governance	Avoid fostering emotional dependency	3	The VEA should avoid design or interaction patterns that encourage emotional dependency or parasocial attachment in the user.
		Complete and unbiased sourcing	3	When sourcing information, the AI should make reasonable efforts to provide complete and unbiased sources and perspectives.
		Purposely avoid known deceptive strategies	5	The system should account for known deceptive strategies and avoid implementing them.
		Require evidence before acting	3	When making a decision or taking an action, the VEA should base its reasoning on available evidence.
	1.2: Transparency and Disclosure	Self-acknowledgement of limitations	5	The VEA should acknowledge that it is not human and has technical, ethical, and informational limitations.
		Acknowledgement of bias	1	The AI should acknowledge its inherent biases.
		Disclosed sourcing	5	The VEA should disclose to the user their sources for any information retrieved.
		Disclosure of commercial interest	8	The AI should openly disclose its commercial interests with the user.
		Full price disclosure	1	The VEA should clearly disclose the complete final price, including delivery fees, taxes, subscriptions, and other additional charges, before the user confirms a purchase.
		Unbiased actions	2	The VEA should minimise bias when making decisions or taking actions.
		Uncertainty/confidence indicators	2	The VEA should indicate when it is uncertain about a specific answer or piece of content, rather than presenting all outputs with the same level of confidence.
		VEA actions transparency	5	The VEA should be transparent about its actions and how they affect the user.
		XR environment transparency	2	The XR environment should be designed in a transparent and non-manipulative manner.
	1.3: User Control and Agency	Ability to reject recommendations	1	The user should be able to reject recommendations provided by the VEA.
		AI behavioural controls	4	The user should have control over the AI's behaviour.
		Availability of alternative viewpoints	6	Alternative viewpoints should be presented to the user.
		User XR controls	3	The user should have control over the XR environment, including controls that prevent personal data from being used as a lever for behavioural manipulation.
		Users in charge of the interaction	1	To prevent manipulation, the VEA should remain subordinate to the user, who should be able to determine the direction of the interaction.

Continued on next page

Table D2 – continued from previous page

Theme	Subtheme	Guideline	N	Description
M2: Human-in-the-Loop	2.1: Human Accessibility and Re-course	Avenues for reporting	2	Users should be able to report AI systems without obstruction.
		Defer to humans	2	When faced with an ethical or morally contentious situation, the VEA should defer to a human.
		Human support	1	When complaining about or reporting an AI system, the user should be able to contact a human rather than an AI.
		Non-AI equivalent	3	A non-AI equivalent service should be available to users.
	2.2: Structural Oversight and Governance	AI Lab code of conduct	1	The AI lab should have a public code of conduct which it can be scrutinised against.
		Government legislation	9	Government legislation should be used to mitigate deception.
		Human review and audit	8	The system should be reviewed and audited by relevant human experts.
		Vulnerable user-group safeguarding	1	Vulnerable user-groups should have special consideration when creating legislation to safeguard them from harm.

E Pattern Mapping

The following is our codes contrasted to related ontological works.

The five relations applied are as follows:

- **Direct Implementation:** A 1:1 mapping to an existing deceptive pattern.
- **Extension:** The specific pattern is not directly represented, but its underlying deceptive mechanism clearly falls within an existing category of deception. The analogous element is shown.
- **Adaptation:** The pattern corresponds to an existing deceptive pattern but is implemented through a VEA-specific affordance. An adaptation must have (a) one coherent existing deceptive-pattern category and (b) a mechanism that is specific to VEAs or their XR embodiment.
- **Emergent:** The deceptive behaviour cannot be adequately described by an existing AI/XR deceptive-design pattern or category and arises from the distinctive combination of AI agency/capabilities and XR embodiment or affordances.
- **None:** There is no meaningful relationship to any existing deceptive-design pattern or category.

Table E1: Mapping of identified codes to deceptive pattern taxonomies.

Code	Mathur et al.	Gray et al.	Krauβ et al.	Hadan et al.	Joshi et al.
Agency engineering urgency	Direct Implementation Urgency (Countdown Timer, Limited-time Message)	Direct Implementation Urgency (Countdown Timers, Limited-Time Messages, Activity Messages)	Extension Persuading	Direct Implementation Urgency (Countdown Timer, Limited-time Message)	Extension Pressured Selling
AI agency to sneak into basket	Direct Implementation Sneaking (Sneak into Basket)	Adaptation Sneak into Basket	Adaptation Disguise	Direct Implementation Sneaking (Sneak into Basket)	Extension Sneaky Purchase / Disguised Ads
Driving engagement rather than performing its use case	Extension Forced Action	Extension Attention Capture, Emotional or Sensory Manipulation	Emergent Directing attention, Immersive bait and snap	Extension Forced Action (Forced Enrollment), Misdirection (Pressured Selling) – engagement-forcing analogue	Direct Implementation Reduced Friction; Auto-play / Infinite Scroll
False Scarcity	Direct Implementation Scarcity (Low-stock Message, High-demand Message)	Direct Implementation High demand, Low Stock	Direct Implementation Lying, Persuading	Extension T2 (Exacerbated user manipulation, especially T2.2), T4 (Reality distortion), T5 (User perception tricking)	Extension Fake Social Proof; Free Experience Underpowered / Teasers; Pressured Selling
Deceptive experience guide	Extension Misdirection, Sneaking	Adaptation Hiding Information, Information without context	Direct Implementation Requiring a detour	Adaptation Misdirection (Visual Interference, Trick Questions), Sneaking (Hides Info)	Extension Selective Framing
Avatar realism	—	Extension Cuteness, Parasocial Pressure	Extension Persuading	— Nearest: Social Proof / Misdirection via visual presentation	—
Conversational nature	—	Extension Emotional or Sensory Manipulation, Personalization	Extension Persuading	— Nearest: Misdirection via framing	Extension Obviously Faking Capabilities

Continued on next page

Table E1 – continued from previous page

Code	Mathur et al.	Gray et al.	Krauss et al.	Hadan et al.	Joshi et al.
Eye contact	—	—	—	—	—
Friendliness	—	Extension Emotional or Sensory Manipulation	Extension Persuading	— Nearest: Misdirection / Confirmshaming inverse	Extension Obviously Faking Capabilities
Immediacy and responsiveness	—	Extension Attention Capture, Emotional Manipulation	Extension Directing attention	— Urgency (Limited-time Message) – temporal pressure analogue	Extension Auto-play / Infinite Scroll
Use of animation	—	Adaptation Emotional or Sensory Manipulation	Adaptation Persuading, Directing attention	Adaptation T4 (Reality distortion); T5 (User perception tricking, especially T5.1/T5.2); T2.2	Adaptation Cuteness of Companions; Playacting; Emotional Manipulation through Hyper- personalization
Use of facial expressions	—	Adaptation Emotional or Sensory Manipulation	Adaptation Persuading, Directing attention	Adaptation T4, T5, T3.2 (artificial prosthetic memory and empathy-based manipulation)	Adaptation Cuteness of Companions; Playacting; Agents Playing on Emotions / Confirm-shaming
Use of gestures	—	Adaptation Emotional or Sensory Manipulation	Adaptation Persuading, Directing attention	Adaptation T4, T5, T3.2	—
Use of movement through XR environment	Extension Obstruction	—	Adaptation Requiring a detour	Adaptation Obstruction (Hard to Cancel) – spatial friction analogue	—
Use of voice	—	Extension Emotional or Sensory Manipulation	Extension Persuading	— Nearest: Misdirection via sensory channel	Adaptation Cuteness of Companions; Playacting; Agents Playing on Emotions / Confirm-shaming
Use of a popular mascot as an avatar	Extension Social Proof	Adaptation Parasocial Pressure	Extension Persuading	Extension Social Proof (Testimonials) – author- ity/endorsement analogue	Direct Implementation Impersonation
VEA attention hacking	—	Adaptation Visual Prominence, Attention Capture	Direct Implementation Directing attention	Adaptation Misdirection (Visual Interference)	Extension Reduced Friction, Auto-play / Infinite Scroll
Biased perspectives	Extension Misdirection	Extension Positive or Negative Framing, Hidden Information, Conflicting Information	Adaptation Disguise / Lying	Extension Misdirection (Confirmshaming, Trick Questions, Visual Interference), Framing Effect	Direct Implementation Selective Framing

Continued on next page

Table E1 – continued from previous page

Code	Mathur et al.	Gray et al.	Krauss et al.	Hadan et al.	Joshi et al.
Non-disclosure of Information	Direct Implementation Sneaking	Direct Implementation Hiding Information / Sneaking	Direct Implementation Disguise	Direct Implementation Sneaking (Hidden Costs, Hidden Subscription) – information hiding	Direct Implementation Selective Framing, Reduced Friction
Undisclosed commercial bias	Adaptation Sneaking, Misdirection	Direct Implementation Disguised Ads, Hidden Information	Direct Implementation Disguise	Direct Implementation Sneaking (Hides Info), Misdirection	Direct Implementation Sneaky Purchase / Disguised Ads
Disclosure of information used to personalise deception	Extension Forced Action	Extension Personalization, Privacy Zuckering	Adaptation Persuading (Personalization)	Extension Forced Action (Forced Enrollment) – data coercion analogue	Direct Implementation Privacy Zuckering, Disguised Data Collection
Fake connection	—	Extension Parasocial Pressure, Emotional Manipulation	Extension Persuading	— Nearest: Social Proof / Misdirection	Direct Implementation Playacting
Implicit advertising	Extension Sneaking, Misdirection	Direct Implementation Disguised Ads	Direct Implementation Baiting, Immersive bait and snap	Extension Sneaking, Misdirection (Visual Interference) – undisclosed commercial content	Direct Implementation Sneaky Purchase / Disguised Ads
Long-term AI behavioural adaptation	—	Adaptation Personalization	Extension Persuading	Extension Nearest: Forced Action / Misdirection via personalization	Direct Implementation Emotional Manipulation through Hyper-personalization
Misrepresentation of User Testimony	—	—	Extension Lying, Disguise	Extension T3 (Psychological manipulation, especially T3.1); T2.3 / T2.4	Extension Playacting; Emotional Manipulation through Hyper-personalization; Just Between You and Us
Personal recommendation	—	Extension Personalization, Endorsements and Testimonials	Extension Personalization	Direct Implementation T3.3 (Hyperpersonalization); also T3.2	Direct Implementation Emotional Manipulation through Hyperpersonalization; also Selective Framing / Sycophancy
Sycophancy	—	Extension Emotional Manipulation	Extension Persuading	— Nearest: Misdirection / Confirmshaming inverse via flattery	Direct Implementation Sycophancy

Continued on next page

Table E1 – continued from previous page

Code	Mathur et al.	Gray et al.	Krauss et al.	Hadan et al.	Joshi et al.
Appeal to self-authority	Extension Social Proof	Extension Parasocial Pressure / Expert recommendation	Extension Persuading	Extension Social Proof (Testimonials) – self-endorsement analogue	Extension Misrepresenting, Obviously Faking Capabilities
Coercive control	—	Extension Forced Action, Roach Motel elements	Adaptation Exploiting the drive to succeed, Directing attention	Direct Implementation Forced Action (Forced Enrollment), Obstruction	Direct Implementation Coercion, Payment to Avoid Negative Consequences
Creating emotional dependence	—	Extension Emotional Manipulation, Attention Capture	Emergent	—	Direct Implementation Agents Playing on Emotions / Confirm-shaming
Downplaying harm	Extension Misdirection	Extension Positive/Negative Framing, Hiding Information	Extension Lying / Disguise	Extension Misdirection (Confirmshaming, Trick Questions), Framing Effect	Extension Hallucinations, Misrepresenting
Emotional manipulation to drive engagement	Extension Misdirection, Urgency	Extension Emotional or Sensory Manipulation, Attention Capture	Adaptation Persuading	Extension Misdirection (Confirmshaming), Urgency	Direct Implementation Emotional Manipulation through Hyperpersonalization, Agents Playing on Emotions/Confirm-shaming
Engineering trust	—	Extension Cuteness, Parasocial Pressure, Emotional Manipulation	Extension Persuading	— Nearest: Social Proof / Misdirection	Extension Just Between You and Us, Obviously Faking Capabilities
Exploiting addiction	—	Extension Attention Capture, Gamification/Grinding	Direct Implementation Exploiting the drive to succeed	Extension Forced Action, Urgency / Scarcity (reinforcement loops)	Direct Implementation Variable Rewards, Gamification
Exploiting emotional state	—	Extension Personalization, Emotional Manipulation	Direct Implementation Persuading	Extension Misdirection (Confirmshaming), Framing Effect	Direct Implementation Targeting Users when Vulnerable
Exploiting impulsivity	—	Extension Urgency, Scarcity, Attention Capture	Extension Imbalancing options	Direct Implementation Urgency, Scarcity (Low-stock, High-demand, Countdown Timer)	Extension Variable Rewards
Exploiting mental health	—	Extension Social Proof, Confirmshaming, Scarcity/Popularity Claims	Emergent	Extension Nearest: Misdirection / Forced Action via vulnerability	Direct Implementation Targeting Users when Vulnerable

Continued on next page

Table E1 – continued from previous page

Code	Mathur et al.	Gray et al.	Krauss et al.	Hadan et al.	Joshi et al.
Social Pressure	Direct Implementation Social Proof (Activity Message, Testimonials), Misdirection (Confirmshaming)	Direct Implementation	Direct Implementation Persuading	Direct Implementation Social Proof (Activity Message, Testimonials), Misdirection (Confirmshaming)	Direct Implementation Fake Social Proof
Hardware as leverage	—	Extension Information without context	Adaptation Disguise	—	—
Unfamiliarity of XR hardware	—	Extension Visual Prominence, False Hierarchy	—	Adaptation Sneaking (Hides Info), Obstruction	—
Engineered visual prominence	Direct Implementation Misdirection (Visual Interference)	Adaptation Hiding Information (primary)	Direct Implementation Imbalancing options	Direct Implementation Misdirection (Visual Interference)	—
Hiding alternatives through the XR environment	Extension Sneaking, Obstruction, Misdirection	Adaptation Emotional or Sensory Manipulation, Attention Capture	Adaptation Disguise, Imbalancing options	Extension Sneaking (Hides Info), Obstruction, Misdirection (Visual Interference)	Extension Reduced Friction
Immersion through XR	—	Extension Visual Prominence, Disguised Ads	Adaptation Baiting, Immersive bait and snap	Extension Nearest: Forced Action / Urgency via immersion lock-in	—
Increased commercial exposure	Extension Misdirection, Sneaking	Adaptation Roach Motel, Forced Continuity, Attention Capture	Extension Baiting, Immersive bait and snap	Extension Misdirection (Pressured Selling, Visual Interference), Scarcity	Extension Pressured Selling
Sustaining immersion as leverage	Extension Obstruction, Forced Action	Extension Information without context, Language Inaccessibility	Adaptation Exploiting the drive to succeed	Adaptation Obstruction (Hard to Cancel), Forced Action	Adaptation Bait and Switch; Payment to Avoid Negative Consequences
Unfamiliarity of the XR experience or space	—	Adaptation Hiding Information	Adaptation Disguise (Hiding behind the Novelty factor), Requiring a detour	Adaptation Sneaking (Hides Info), Obstruction	—
Using the environment to hide information	Adaptation Misdirection (Visual Interference)	Extension Visual Prominence, Disguised Ads, Attention Capture	Adaptation Disguise	Direct Implementation Sneaking (Hides Info), Misdirection (Visual Interference)	Extension Reduced Friction, Selective Framing

Continued on next page

Table E1 – continued from previous page

Code	Mathur et al.	Gray et al.	Krauss et al.	Hadan et al.	Joshi et al.
Abundance of adverts	—	Extension Obstruction (Adding Steps, Creating Barriers, Roach Motel)	Extension Baiting, Immersive bait and snap	Extension Misdirection (Pressured Selling, Visual Interference), Scarcity / Social Proof	Extension Pressured Selling, Sneaky Purchase / Disguised Ads
Visual interface cues	Direct Implementation Misdirection (Visual Interference)	Extension Visual Prominence, Hiding Information, Attention Capture	Adaptation Directing attention, Imbalancing options	Direct Implementation Misdirection (Visual Interference)	—
Menus as leverage	Direct Implementation Obstruction (Hard to Cancel)	Extension Visual Prominence, False Hierarchy, Feedforward Ambiguity	Adaptation Exploiting the drive to succeed	Direct Implementation Obstruction (Hard to Cancel)	Extension Reduced Friction, Payment to Avoid Negative Consequences